Multimodal Prediction of Conversational Dynamics: Turn-Taking, Backchanneling, and Listening using Text, Audio, and Visual Signals

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by

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May, 2026

Declaration

This is to certify that the thesis entitled "**Multimodal Prediction of Conversational Dynamics: Turn-Taking, Backchanneling, and Listening using Text, Audio, and Visual Signals**", submitted by me to the *Indian Institute of Technology Ropar*, for the award of the degree of Master of Technology, is a bonafide work carried out by me under the supervision of **Dr. Mudasir Ahmad Ganaie**. The content of this thesis, in full or in parts, have not been submitted to any other University or Institute for the award of any degree or diploma. I also wish to state that to the best of my knowledge and understanding nothing in this report amounts to plagiarism.

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Sincerely
Ankit Navrang Meshram

Certificate

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Abstract

Human conversation relies on intricate, sub-second coordination through mechanisms like turn-taking and backchanneling. While modern spoken dialogue systems predominantly rely on simplistic silence-thresholds, natural dialogue fundamentally depends on predicting transitions using parallel linguistic, acoustic, and visual cues. To close this gap, AI systems must effectively fuse multimodal signals in real time. However, achieving this integration efficiently and robustly for real-world deployment remains a significant challenge.

This thesis presents a comprehensive evaluation of multimodal fusion mechanisms for predicting three mutually exclusive listener actions: TURN, KEEP, and BACKCHANNEL, utilizing the tri-modal Multi-Modal Face-to-Face (MM-F2F) dataset. Through a systematic benchmark of eleven diverse architectures, this work identifies a fundamental limitation in the existing literature, termed the **positive-correlation ceiling**. Existing fusion paradigms—whether based on Hadamard products, attention, or weighted averaging—are designed to amplify cross-modal agreement, effectively discarding the informative directional conflict between modalities. Furthermore, the analysis reveals critical vulnerabilities in highly efficient baseline models; notably, Low-Rank Multimodal Fusion (LMF) suffers catastrophic failure under missing-modality conditions, rendering it unsuitable for deployment where sensor dropout is possible.

To address these limitations, this thesis proposes two novel architectures: **Anti-Correlation Gated Fusion (ACGF)** and the **Tiny Anti-Correlator (TAC)**. Both models explicitly encode cross-modal conflict alongside agreement using signed difference vectors. TAC, pushing the limits of parameter efficiency, replaces learned linear projections with parameter-free element-wise operations and a minimal gating network.

Experimental results demonstrate that predictive performance is largely decoupled from fusion-module complexity when modality encoders are frozen. The proposed TAC architecture achieves a Macro-F1 score of 0.8523 using only **5,173 fusion parameters**—performing within 0.84% of the top-performing Gated Multimodal Unit (GMU) baseline while requiring 159 times fewer parameters. Crucially, the explicit conflict-modeling pathway enables both TAC and ACGF to degrade gracefully under simulated sensor dropout, maintaining informative representations where traditional Hadamard-based methods collapse. Ultimately, this thesis demonstrates that highly parameter-efficient, robust, and interpretable multimodal fusion is achievable, establishing a new practical frontier for deployable, full-duplex conversational AI.

Contents

Acknowledgement iii

Certificate iv

List of Figures xi

List of Tables xiii

1 Introduction 1

1.1 Background and Motivation . . . 1
1.2 The Multimodal Nature of Conversational Dynamics . . . 3
1.3 Predicting Conversational Dynamics: Problem Formulation . . . 4
1.4 The Fusion Benchmark and Parameter Efficiency . . . 5
1.5 The Positive-Correlation Ceiling and Proposed Solution . . . 6
1.6 Thesis Objectives . . . 7
1.7 Thesis Organisation . . . 7

2 Literature Review 9

2.1 Turn-Taking and Backchannel Prediction . . . 9
2.1.1 Theoretical Foundations . . . 9
2.1.2 Text-Based Approaches . . . 9
2.1.3 Acoustic Approaches . . . 10
2.1.4 Multimodal and Visual Approaches . . . 10
2.1.5 Backchannel Prediction . . . 11
2.2 Multimodal Fusion Strategies . . . 11
2.2.1 Early and Late Fusion . . . 11
2.2.2 Tensor-Based Fusion . . . 12
2.2.3 Gating-Based Fusion . . . 13
2.2.4 Attention-Based Fusion . . . 13
2.2.5 The Shared Limitation: Positive-Correlation Fusion . . . 14
2.3 Parameter-Efficient Architectures . . . 14
2.3.1 Efficiency in Unimodal Models . . . 15
2.3.2 Lightweight Fusion Modules . . . 15
2.3.3 Robustness Under Modality Dropout . . . 15
2.4 Datasets for Conversational Dynamics . . . 16
2.4.1 Text-Only Corpora . . . 16
2.4.2 Multimodal Corpora . . . 16
2.5 Research Gaps and Positioning of This Thesis . . . 17

3 Methodology **19**
3.1 Dataset . . . 19
3.2 Uni-Modal Encoder Backbones . . . 19
3.2.1 Text Encoder: GPT-2 . . . 19
3.2.2 Audio Encoder: HuBERT . . . 20
3.2.3 Video Encoder: VideoMAE . . . 20
3.2.4 Shared Feature Space . . . 20
3.3 Two-Stage Training Framework . . . 21
3.3.1 Stage 1: Uni-Modal Encoder Training . . . 21
3.3.2 Stage 2: Fusion Module Training . . . 21
3.4 Baseline Fusion Mechanisms . . . 21
3.4.1 Early Fusion . . . 23
3.4.2 Late Fusion . . . 23
3.4.3 Tensor Fusion Network (TFN) . . . 24
3.4.4 Low-Rank Multimodal Fusion (LMF) . . . 24
3.4.5 Tucker Fusion . . . 25
3.4.6 Gated Multimodal Unit (GMU) . . . 25
3.4.7 Multimodal Factorised Bilinear Pooling (MFB) . . . 26
3.4.8 Cross-Modal Attention . . . 26
3.4.9 Multimodal Transformer (MulT) . . . 27
3.4.10 Bi-Bimodal Fusion Network (BBFN) . . . 27
3.4.11 Quaternion Fusion . . . 28
3.5 Proposed Fusion Architectures . . . 28
3.5.1 Signed Difference Vectors . . . 29
3.5.2 Anti-Correlation Gated Fusion (ACGF) . . . 29
3.5.3 Tiny Anti-Correlator (TAC) . . . 31
3.6 Training Configuration and Evaluation Methodology . . . 32
3.6.1 Optimiser and Hyperparameters . . . 32
3.6.2 Loss Function . . . 33
3.6.3 Ablation Study Design . . . 33
3.6.4 Evaluation Metrics . . . 33
3.6.5 Parameter Count Reporting . . . 34

4 Experimental Results and Analysis **35**
4.1 Uni-Modal Baseline Results . . . 35
4.2 Bi-Modal Fusion Results . . . 35
4.3 Full Tri-Modal Benchmark Results . . . 36
4.4 Parameter Efficiency Analysis . . . 36
4.5 Ablation Study: Robustness Under Missing Modalities . . . 38
4.6 The Positive-Correlation Ceiling . . . 39
4.7 Summary of Findings . . . 41

5 Conclusion **43**

5.1 Overview . . . 43

5.2 What This Thesis Achieved . . . 43

5.3 Broader Significance . . . 44

5.4 Future Directions . . . 45

5.5 Closing Remarks . . . 45

References **47**

List of Figures

1.1 Reactive vs. Predictive conversational systems. 2
1.2 The dynamics of human conversational turn-taking. 2
1.3 The tri-modal, three-class problem formulation for conversational dynamics. 4
1.4 Resolving multimodal conflict with Signed Difference Vectors. 7

2.1 A representative face-to-face dyadic interaction from the MM-F2F dataset. 17

3.1 Two-stage training framework. 22
3.2 Early Fusion 23
3.3 Late Fusion 23
3.4 Tensor Fusion Network 24
3.5 Low-Rank Multimodal Fusion 24
3.6 Tucker Fusion 25
3.7 Gated Multimodal Fusion 25
3.8 Multimodal Factorise Bilinear Pooling 26
3.9 Cross-Modal Attention Fusion 26
3.10 Multimodal Transformer 27
3.11 Bi-Bimodal Fusion Network 27
3.12 Quternion Fusion 28
3.13 Anti-Correlation Gated Fusion 29
3.14 Tiny Anti-Correlator 31

4.1 Fusion parameters (log scale) vs. accuracy in the full tri-modal condition. . 37
4.2 Mac-F1 across all ablation conditions for all eleven models. 39
4.3 Distribution of accuracy across all runs (full tri-modal + ablation conditions, all models). 40

List of Tables

3.1 ACGF parameter breakdown ($d = 256$). 30
3.2 TAC parameter breakdown ($d = 256$). 32
3.3 Fusion module parameter counts for all evaluated methods, ordered by parameter count. 34

4.1 Stage 1 uni-modal baselines. Bold = best per column. 35
4.2 Bi-modal GMU results on the MM-F2F test set. 36
4.3 Full tri-modal benchmark, ranked by Mac-F1. 37
4.4 Parameter efficiency summary. 38
4.5 Ablation study: Mac-F1 under missing modality conditions. 38

Chapter 1

Introduction

1.1 Background and Motivation

Human conversation is one of the most intricate forms of social behaviour that we engage in every day. Unlike written communication, spoken dialogue is inherently dynamic, multi-layered, and deeply time-sensitive. When two people speak face to face, they are not merely exchanging words — they are simultaneously decoding prosodic signals, interpreting facial expressions, tracking gaze direction, and anticipating each other's next move, all within fractions of a second. This remarkable coordination, which most humans perform without conscious effort, represents one of the grand unsolved challenges in the field of artificial intelligence and human-computer interaction.

At the heart of this challenge lies the problem of **conversational dynamics** — the mechanisms by which participants in a conversation decide when to speak, when to listen, and when to provide brief signals of engagement. Two of the most fundamental and frequent of these mechanisms are **turn-taking** and **backchanneling**. Turn-taking refers to the process by which speakers coordinate the conversational floor — knowing when to begin speaking, when to yield, and when a transition is appropriate. Backchanneling refers to the short, overlapping responses that listeners produce — sounds like "uh-huh", "right", "I see", or nodding — that signal active engagement without taking the conversational floor. Together, these two mechanisms form the scaffolding of natural human dialogue.

Despite decades of research in speech and language processing, modern spoken dialogue systems remain deeply inadequate in replicating these mechanisms. The vast majority of commercial voice assistants — whether embedded in smartphones, smart speakers, or customer service applications — rely on a profoundly simplistic model of turn management: they detect silence, and when silence exceeds a predefined threshold (typically around 700 milliseconds), they assume the user has finished speaking and begin formulating a response [1].

This approach leads to two well-known failure modes: either the system interrupts the user mid-thought by treating a brief pause as a turn completion, or it responds with a noticeable delay that makes the interaction feel sluggish and unnatural. Neither failure is acceptable in applications where smooth, responsive dialogue is critical.

What makes this problem particularly difficult is that silence is only one of many cues that human listeners use to coordinate conversation. Research in linguistics and

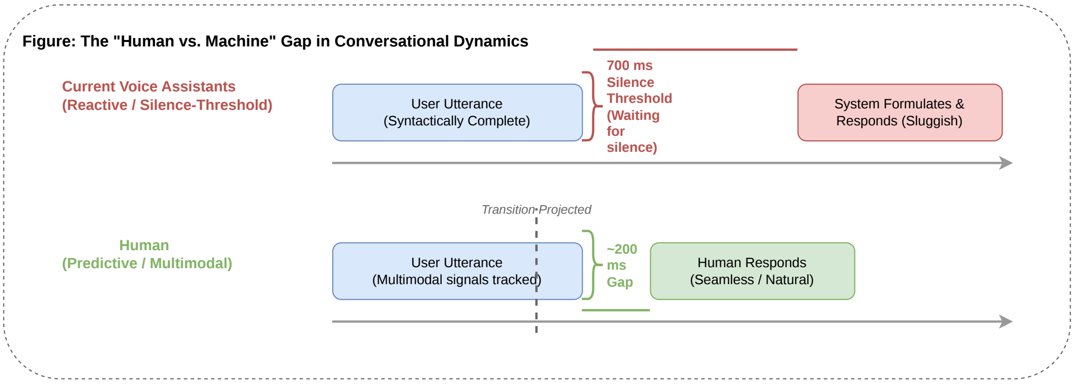

Figure 1.1: A comparison of conversational floor management. Traditional voice assistants rely on a reactive silence-threshold (top), leading to unnatural delays. Human listeners track multimodal cues in parallel, allowing them to project transitions and respond within the natural ∼200 ms timeframe.

psycholinguistics has established that turn-taking is, in fact, a highly predictive process. Listeners begin preparing their responses before the speaker has finished speaking, relying on a rich constellation of signals that include syntactic and pragmatic completeness of the utterance, prosodic patterns such as pitch declination and rhythm, visual cues such as gaze direction and head nods, and the broader context of the ongoing conversation [2, 3, 4]. The typical gap between turns in natural human conversation is remarkably small — often around 200 milliseconds [4] — far too short for a listener who is merely waiting for silence to react, prepare, and begin speaking. This implies that listeners are continuously projecting and anticipating turn completions in real time, using all available modalities simultaneously.

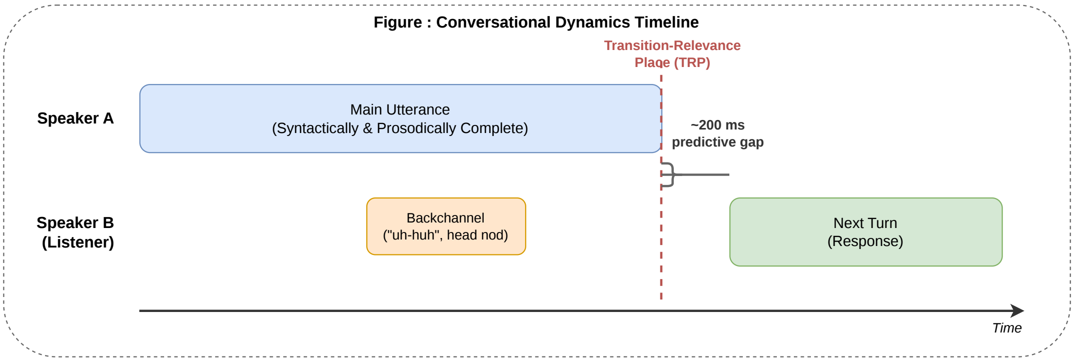

Figure 1.2: The dynamics of human conversational turn-taking. Listeners actively project Transition-Relevance Places (TRPs) and prepare responses before the current speaker finishes, resulting in rapid conversational floor transfers (typically around 200 ms). Backchannels occur concurrently with the main utterance to signal active engagement without triggering a turn shift.

Backchanneling presents an equally complex challenge. Unlike turn-taking, which involves a transfer of the conversational floor, a backchannel is produced precisely to avoid such a transfer — it signals that the listener is engaged and wishes the speaker to continue. The decision to produce a backchannel, and which type to produce, is driven not primarily by linguistic content but by prosodic and visual cues: a rising intonation, a prolonged gaze, or a brief pause that invites acknowledgment [3]. A dialogue system that processes only text is fundamentally blind to these signals and will systematically fail to produce backchannels at the right moments, making the interaction feel one-sided and robotic.

The gap between how humans naturally converse and how current dialogue systems behave is therefore not merely a matter of engineering refinement — it reflects a fundamental architectural limitation. To close this gap, a dialogue system must be capable of processing linguistic, acoustic, and visual signals in parallel, integrating them in real time to predict conversational events as they unfold.

1.2 The Multimodal Nature of Conversational Dynamics

A key insight that motivates this thesis is that no single modality is sufficient for predicting conversational dynamics reliably. Text carries semantic and syntactic information that is crucial for projecting pragmatic completion — whether an utterance has reached a point where a turn transition would be appropriate [1]. However, text alone is blind to prosodic emphasis, which can fundamentally change the communicative intent of an utterance, and blind to visual signals, which are known to play a significant role in floor management [3].

Audio carries acoustic information — pitch trajectories, speaking rate, voice quality, and inter-pausal structure — that is tightly coupled with the production and perception of turn-taking cues. Research by Ekstedt and Skantze [5] has shown that acoustic models alone can achieve meaningful performance on turn-taking prediction precisely because prosody encodes syntactic and pragmatic structure implicitly. However, audio-only models lack the semantic grounding that language provides and are sensitive to noise and channel variation.

Visual signals, particularly facial expressions, gaze direction, and head movements, have been identified as important modulators of conversational behaviour since the foundational work of Kendon [3]. Speakers tend to avert their gaze during an utterance and shift their gaze toward the listener when approaching a turn-completion point. Head nods from the listener signal that a backchannel is forthcoming. These cues are largely independent of linguistic content and provide complementary information that neither text nor audio alone can replicate.

The implication is clear: a system that can integrate all three modalities — text, audio, and visual — should, in principle, be able to predict conversational dynamics more accurately than any unimodal or bimodal system [6, 7]. However, achieving this integration is non-trivial. Each modality operates on a different timescale, has a different representation space, and contributes different information. The question of how to fuse these modalities effectively — how to align them, how to weight them, and how to handle cases where they provide conflicting signals — is the central technical challenge addressed by this thesis.

Equally important, and often overlooked in the research literature, is the question of

parameter efficiency. Large-scale multimodal models that achieve strong performance often do so at enormous computational cost, requiring hundreds of millions of parameters and significant inference time [8]. For real-world deployment in full-duplex dialogue systems — where predictions must be made continuously and at low latency, often on devices with constrained resources — parameter efficiency is not a luxury but a strict requirement. This thesis therefore pays explicit attention to the trade-off between prediction accuracy and model size, seeking fusion architectures that achieve competitive performance without unnecessary computational overhead.

1.3 Predicting Conversational Dynamics: Problem Formulation

This thesis addresses the problem of predicting all three primary listener actions — **turn-taking**, **backchanneling**, and **keeping** (continuing to listen silently) — from a multimodal signal comprising text, audio, and video. These three actions form a complete and mutually exclusive partition of the listener's response space at any given moment [6]. Formally, given a word utterance within an ongoing conversation, the model must assign one of three labels:

- **KEEP**: the current speaker continues; the system remains silent.
- **TURN**: a turn shift occurs; the system takes the conversational floor [2].
- **BACKCHANNEL (BC)**: the listener produces brief feedback (e.g., "mm-hmm") without taking the floor [3].

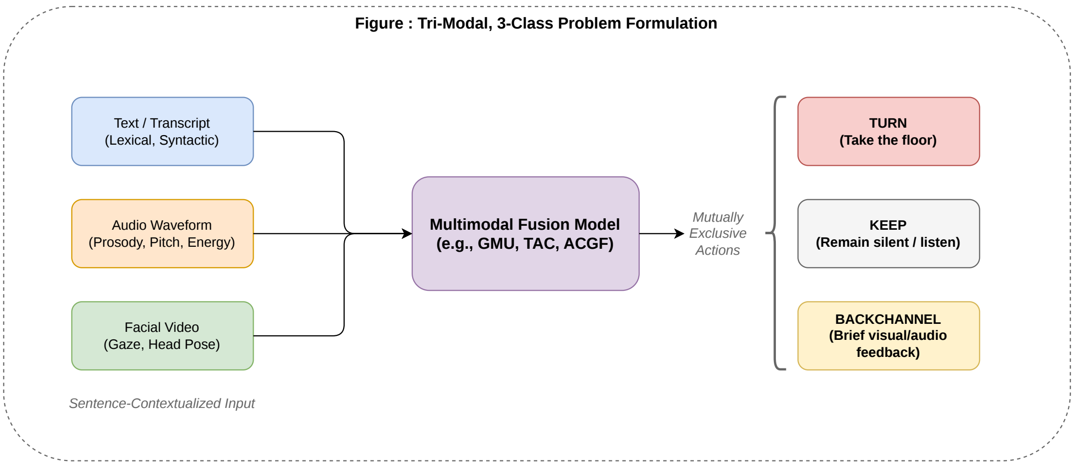

Figure 1.3: The tri-modal, three-class problem formulation for conversational dynamics. Sentence-contextualised linguistic, acoustic, and visual signals are integrated to predict one of three mutually exclusive listener actions at any given moment: taking the floor (TURN), remaining silent (KEEP), or providing brief feedback (BACKCHANNEL)

This is a word-level, sentence-contextualised, three-class classification problem. Predicting which of these labels applies at each moment — and doing so in real time from multimodal input — is the central task of this work.

This tri-modal, three-class formulation is grounded in an important empirical observation: when text, audio, and video signals are all available, the same architectural framework can naturally address all three listener actions within a single unified model, rather than treating turn-taking and backchanneling as separate problems requiring separate systems [6]. This unification is both practically convenient and theoretically motivated — the cues that drive the three actions overlap considerably, and a joint model can exploit these shared representations efficiently. In particular, the boundary between a KEEP and a TURN is heavily conditioned on prosodic completion [1, 4], while the boundary between a KEEP and a BC is conditioned on acoustic and visual acknowledgment cues [3, 9] — making multimodal input essential for disambiguating all three classes simultaneously.

The dataset that makes this formulation empirically tractable is the **Multi-Modal Face-to-Face (MM-F2F)** dataset, introduced by Lin et al. [6]. MM-F2F is, to the best of our knowledge, the first face-to-face human conversation dataset that covers all three modalities — text, audio, and de-identified facial video — and provides word-level annotations for turn-taking, backchanneling, and listening actions. It comprises 773 videos, approximately 210 hours of conversation, 955 unique speakers, and over 169,000 annotated utterances.

1.4 The Fusion Benchmark and Parameter Efficiency

Building on this dataset and problem formulation, this thesis makes a focused and systematic contribution: a **benchmark of multimodal fusion mechanisms** for conversational dynamics prediction, evaluated simultaneously for predictive performance and parameter efficiency.

The central research question is: given that text, audio, and video are each individually informative, what is the best way to combine them — and at what computational cost? This question has both empirical and theoretical dimensions [7, 8]. Different fusion architectures make fundamentally different assumptions about how modalities interact. Early fusion methods concatenate raw feature vectors and learn a joint representation from scratch [10]. Late fusion methods combine the outputs of independently trained modality-specific models [11]. Methods such as Low-rank Multimodal Fusion (LMF) [12], Gated Multimodal Units (GMU) [13], Multimodal Factorised Bilinear pooling (MFB) [14], and the Multimodal Transformer [15] make varying assumptions about the nature of cross-modal interactions — some assuming primarily additive contributions, others allowing for multiplicative interactions or attention-based weighting.

A key finding of this thesis is that **parameter count and predictive performance are not tightly correlated** in this setting. The LMF architecture [12], with approximately 37k parameters in its fusion module, achieves performance comparable to GMU [13], which requires approximately 820k fusion parameters, across tri-modal and bimodal evaluation conditions — a 22-fold difference in parameter count for negligible difference in Macro-F1. This finding has immediate practical implications: it demonstrates that highly parameter-efficient multimodal fusion is achievable for this task.

However, this parameter efficiency analysis also reveals a critical and practically important failure mode. Under missing-modality conditions — where one or two modalities are absent due to sensor dropout, data unavailability, or deployment constraints — LMF's performance collapses catastrophically, falling to Macro-F1 scores between 0.09 and 0.27, which is below random-baseline performance. This collapse is a direct consequence of LMF's Hadamard product design [12]: the element-wise multiplication produces a zero representation whenever any modality is absent, destroying all information carried by the remaining modalities. GMU [13], by contrast, degrades gracefully under the same conditions, retaining Macro-F1 scores between 0.47 and 0.78. This observation motivates the development of a new fusion mechanism that is parameter-efficient while maintaining robust performance under modality dropout.

1.5 The Positive-Correlation Ceiling and Proposed Solution

Beyond the parameter efficiency analysis, this thesis identifies a deeper theoretical limitation that is shared by all evaluated fusion mechanisms — a limitation we term the **positive-correlation ceiling**. All of the architectures benchmarked in this work — including GMU [13], LMF [12], Early Fusion [10], Late Fusion [11], MFB [14], Cross-Attention [16], and the Multimodal Transformer [15] — exploit only positive correlations between modalities. That is, they amplify signal when modalities agree and produce weighted averages of modality representations regardless of whether those modalities are providing consistent or contradictory predictions.

This design assumption is problematic for conversational dynamics prediction because **cross-modal conflict is itself informative**. Consider a moment where the linguistic content of an utterance is syntactically complete [1], suggesting a TURN transition is imminent, but the speaker's prosodic contour is rising and their gaze is sustained [3], suggesting they intend to continue speaking. This conflict between modalities is precisely the kind of ambiguous transition that is most difficult to classify and most consequential to classify correctly. Yet, in all evaluated architectures, this conflicting signal is averaged away rather than being represented and exploited explicitly.

This thesis proposes a principled solution to this problem: **Signed Difference Vectors** — a representation-level approach that encodes cross-modal conflict explicitly by computing pairwise differences between modality feature vectors and concatenating these conflict signals with the original features as input to the prediction head. This approach adds no additional parameters and can be incorporated into any existing fusion architecture as a zero-cost upgrade. The motivation is grounded in the negative-correlation learning literature [17], which demonstrates that explicitly modelling disagreement between

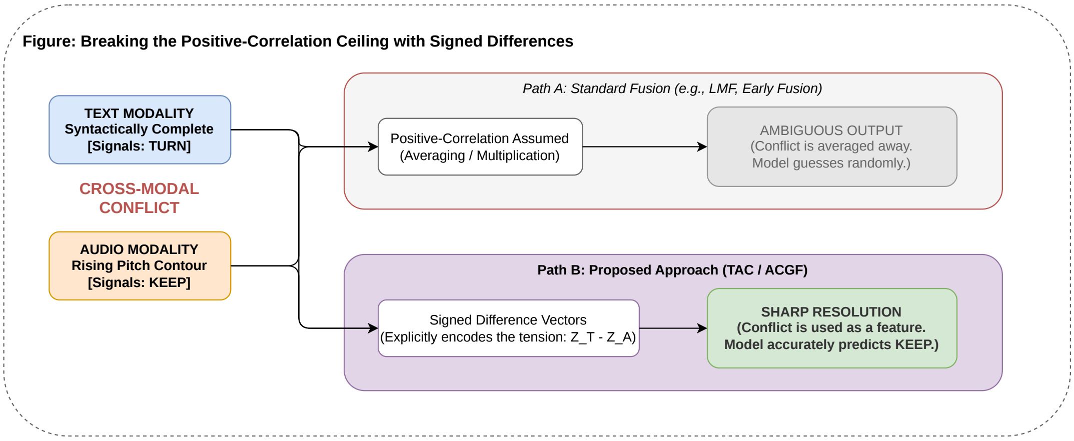

Figure 1.4: The positive-correlation ceiling in action. When modalities provide conflicting signals at ambiguous transitions, standard fusion architectures (Path A) average the conflict, resulting in ambiguous outputs. The proposed methods (Path B) utilize Signed Difference Vectors to explicitly encode cross-modal tension, turning conflict into an informative feature for sharp resolution.

ensemble members improves overall ensemble accuracy by ensuring each member captures complementary information.

1.6 Thesis Objectives

The specific objectives of this thesis are as follows:

Objective 1: To design a controlled benchmarking framework that enables fair and systematic comparison of multimodal fusion mechanisms [7] by controlling for encoder quality, dataset composition, and evaluation methodology.

Objective 2: To benchmark a comprehensive set of fusion mechanisms spanning early [10], late [11], and intermediate fusion paradigms [12, 13, 15, 14, 18, 19, 20], and to identify the best-performing architecture for tri-modal conversational dynamics prediction on the MM-F2F dataset [6].

Objective 3: To identify and evaluate parameter-efficient fusion architectures that achieve predictive performance comparable to much larger models, and to characterise the accuracy-versus-parameter-count trade-off across all evaluated methods with practical deployment implications.

Objective 4: To identify and theoretically characterise the positive-correlation ceiling as a fundamental barrier shared by all evaluated fusion mechanisms, and to propose parameter-free solutions based on signed difference vectors, motivated by negative correlation learning [17].

1.7 Thesis Organisation

The remainder of this thesis is organised as follows:

Chapter 2 reviews the relevant literature on turn-taking and backchannel prediction [2, 3, 1, 5, 9, 6], multimodal fusion methods [10, 12, 19, 13, 14, 15, 18, 16], parameter-efficient architectures, and the datasets available for this task [21, 6].

Chapter 3 presents the model architecture and fusion benchmark framework, describing each of the fusion mechanisms evaluated, the uni-modal encoder backbones, the parameter counts, and the training and evaluation methodology.

Chapter 4 reports the experimental results across all fusion mechanisms, including uni-modal baselines, bimodal experiments, and the full tri-modal benchmark. It presents the parameter efficiency analysis, the ablation study on modality removal, and the analysis of the positive-correlation ceiling.

Chapter 5 concludes the thesis with a summary of contributions, a discussion of limitations and open questions.

Chapter 2

Literature Review

This chapter situates the thesis within the broader research landscape. Section 2.1 reviews computational approaches to turn-taking and backchannel prediction. Section 2.2 surveys multimodal fusion strategies, with particular attention to the shared limitation that motivates this work. Section 2.3 reviews parameter-efficient architectures for multimodal learning. Section 2.4 describes the datasets that have been used for conversational dynamics research. Section 2.5 identifies the research gaps that this thesis addresses.

2.1 Turn-Taking and Backchannel Prediction

2.1.1 Theoretical Foundations

The scientific study of conversational turn-taking was formalised by Sacks et al. [2], who introduced the concept of **transition-relevance places** (TRPs) — points in an utterance at which a speaker handover becomes interactionally appropriate. Their seminal work demonstrated that turn-taking is not a random or reactive process but a highly structured, projectable phenomenon governed by syntactic, prosodic, and pragmatic cues. This theoretical framework remains the foundational reference for all computational work in the area and directly motivates the three-class formulation adopted in this thesis: TURN (a transition occurs), BC (feedback is produced without transition), and KEEP (the current speaker continues).

Complementary to the linguistic approach, Kendon [3] demonstrated through detailed observational studies that gaze direction plays a systematic role in conversational coordination. Specifically, speakers tend to avert their gaze at the onset of an utterance and shift it back toward the listener when approaching a turn-completion point — a signal that anticipates and coordinates the handover before it occurs. This finding has been replicated and extended by subsequent researchers and provides the theoretical justification for the inclusion of visual signals in multimodal turn-taking models.

2.1.2 Text-Based Approaches

Early computational approaches to turn-taking relied primarily on syntactic features, treating turn completion as a function of grammatical completeness. However, as Sacks et al. [2] noted, syntactic completion is a necessary but not sufficient condition for a TRP — prosodic and pragmatic factors are equally important.

The advent of large pretrained language models significantly changed what was achievable with text alone. Ekstedt and Skantze [1] introduced **TurnGPT**, a GPT-2-based language model fine-tuned for turn-taking prediction on the Switchboard and MAPTASK corpora. TurnGPT substantially outperforms LSTM and POS-bigram baselines by capturing pragmatic, not just syntactic, completeness. Their ablation study showed that each additional preceding turn of context improved balanced accuracy, with the steepest gain between zero and one context turns — a finding that underscores the importance of dialogue-level context over sentence-level modelling and directly motivates the contextualised formulation adopted in this thesis. TurnGPT frames the task as predicting end-of-turn tokens embedded in the dialogue sequence, allowing the model to learn turn-completion as a language modelling objective rather than as an external classification label.

Subsequent text-based work has explored larger pretrained models and more fine-grained dialogue act classification. The Switchboard Dialogue Act [21] corpus remains the standard benchmark for this line of work, providing 1,155 telephone conversations with detailed dialogue act annotations including multiple backchannel subtypes. However, text-only models face an intrinsic ceiling: the prosodic and visual signals that trigger many listener actions — particularly backchannels — are simply not recoverable from transcribed text.

2.1.3 Acoustic Approaches

Motivated by the known importance of prosody in turn-taking, a parallel line of work has focused on acoustic modelling. Lee and Narayanan [9] demonstrated that acoustic features such as pitch, energy, and speaking rate contribute meaningfully to the prediction of interruptions in dyadic spoken interactions, outperforming lexical features alone for certain turn types.

The most significant recent contribution in this direction is the **Voice Activity Projection (VAP)** model introduced by Ekstedt and Skantze [5]. VAP extends turn-taking prediction into the continuous acoustic domain using self-supervised speech representations, framing the task as predicting the future voice activity of both participants in a conversation at every acoustic frame. This continuous-prediction formulation is particularly powerful because it removes the dependency on utterance segmentation and enables real-time, streaming inference — a property that is highly desirable for full-duplex dialogue systems. However, VAP processes audio only and is therefore blind to the visual cues identified by Kendon and others as important complements to acoustic signals.

2.1.4 Multimodal and Visual Approaches

The integration of visual signals into turn-taking prediction has a longer history than might be expected, given the technical challenges involved. Kendon's [3] foundational gaze findings were operationalised computationally in a number of systems that used gaze direction, head pose, and facial action units as features alongside acoustic and lexical inputs. These early systems demonstrated consistent improvements over audio-only baselines but were typically limited to small datasets collected under controlled laboratory conditions and used hand-crafted features rather than learned representations.

The recent availability of large-scale, naturally recorded multimodal conversation datasets has enabled deep learning approaches to this problem. The **MM-F2F** framework introduced by Lin et al. [6] represents the most directly relevant prior work to this thesis. Lin et al. present the first tri-modal face-to-face conversation dataset annotated for both turn-taking and backchannel actions, and evaluate a range of fusion approaches on this dataset. Their reported results show a 10% F1 improvement for turn-taking and 33% for backchannel prediction over the text-only TurnGPT baseline when all three modalities are fused — a compelling quantitative demonstration of the value of multimodal integration for this task. However, Lin et al. do not systematically evaluate the parameter efficiency of the fusion mechanisms they consider, nor do they address the theoretical limitation of positive-correlation fusion that is central to this thesis.

2.1.5 Backchannel Prediction

Backchannel prediction has received somewhat less attention than turn-taking in the computational literature, partly because backchannels are less frequent, more diverse in form, and more acoustically subtle than turn transitions. Early computational work framed backchannel prediction as a binary classification problem — BC versus non-BC — and used prosodic features such as pitch reset, rhythm, and energy trajectory as predictors. More recent work has treated backchannel prediction as a subproblem within a broader conversational dynamics framework, which is the approach adopted in this thesis. The MM-F2F formulation [6] treats backchannel as one of three mutually exclusive listener actions (alongside TURN and KEEP), enabling a unified model to predict all three within a single classification framework. This formulation is theoretically cleaner and practically more useful than separate binary classifiers for each action type, as it forces the model to explicitly trade off between competing hypotheses about what the listener will do next.

2.2 Multimodal Fusion Strategies

Multimodal fusion is the process of combining information from two or more modality-specific representations into a single unified representation for downstream prediction. Fusion methods are broadly categorised as **early** (feature-level), **late** (decision-level), or **intermediate** (model-level), depending on where in the processing pipeline the modalities are combined. This section reviews the principal methods evaluated in this thesis.

2.2.1 Early and Late Fusion

Early fusion concatenates the raw or encoded feature vectors of all modalities before passing them to a shared prediction head [10]. This approach has the advantage of allowing the prediction head to learn arbitrary cross-modal interactions from the concatenated representation, but the disadvantage of producing a very high-dimensional input that can be difficult to optimise, particularly with limited data. Early fusion also has no mechanism for handling missing modalities gracefully: if one modality is absent, its contribution to the concatenated vector is typically replaced with a zero vector, which the prediction head must learn to ignore.

Late fusion trains a separate model for each modality and combines their output predictions — typically by weighted averaging or a learned combiner [11]. This approach is simple and robust to missing modalities, since each modality-specific model can be evaluated independently. However, late fusion prevents the model from learning cross-modal interactions at the feature level, which limits its representational capacity relative to methods that combine modalities before the final prediction.

2.2.2 Tensor-Based Fusion

Tensor Fusion Network (TFN) [19] addresses the representational limitations of early and late fusion by computing the outer product of all modality representations, explicitly modelling unimodal, bimodal, and trimodal interactions. Given modality representations $\mathbf{z}_T$, $\mathbf{z}_A$, $\mathbf{z}_V$, TFN computes:

$$\mathbf{z}_{\text{fused}} = [1; \mathbf{z}_T] \otimes [1; \mathbf{z}_A] \otimes [1; \mathbf{z}_V]$$

where $\otimes$ denotes the outer product and the prepended 1 allows unimodal and bimodal terms to be included in the same tensor. The resulting representation has dimension $(d_T + 1)(d_A + 1)(d_V + 1)$, which grows cubically with the modality dimensionalities — making TFN computationally expensive for high-dimensional representations.

Low-rank Multimodal Fusion (LMF) [12] addresses this scalability problem by decomposing the TFN weight tensor into low-rank factors. Rather than learning a full weight tensor $\mathbf{W} \in \mathbb{R}^{d_T \times d_A \times d_V \times d_{\text{out}}}$, LMF learns modality-specific factors $\mathbf{W}_T$, $\mathbf{W}_A$, $\mathbf{W}_V$ of rank $r \ll d$, dramatically reducing the parameter count while retaining the ability to model multiplicative interactions. The fusion is computed as:

$$\hat{\mathbf{y}} = \sum_{i=1}^{r} \left(\mathbf{W}_T^{(i)} \mathbf{z}_T\right) \odot \left(\mathbf{W}_A^{(i)} \mathbf{z}_A\right) \odot \left(\mathbf{W}_V^{(i)} \mathbf{z}_V\right)$$

where $\odot$ denotes element-wise multiplication (the Hadamard product). This design makes LMF highly parameter-efficient, with fusion modules requiring as few as 37K parameters in the configuration used in this thesis. However, as demonstrated in our experiments, the Hadamard product at the core of LMF means that if any modality representation is a zero vector (as occurs under modality dropout), the entire fused representation collapses to zero — a critical failure mode in real-world deployment.

Tucker Fusion [20] generalises LMF by introducing a core tensor that captures higher-order interactions between the modality factors. While Tucker Fusion is more expressive than LMF, it also has more parameters and is more computationally expensive.

2.2.3 Gating-Based Fusion

Gated Multimodal Unit (GMU) [13] introduces learned gating vectors that control the relative contribution of each modality to the fused representation. Given modality representations, GMU computes transformation and gating vectors for each modality:

$$\mathbf{h}_m = \tanh(\mathbf{W}_m \mathbf{z}_m), \quad \mathbf{g}_m = \sigma(\mathbf{W}_{g,m}[\mathbf{z}_T; \mathbf{z}_A; \mathbf{z}_V])$$

and produces the fused representation as the gated sum:

$$\mathbf{h}_{\text{fused}} = \sum_m \mathbf{g}_m \odot \mathbf{h}_m$$

The gating mechanism allows GMU to learn modality-specific weights that are conditioned on the full multimodal context, giving it substantially more flexibility than early or late fusion. GMU achieves the best performance of all evaluated baselines in our benchmark, with a Macro-F1 of 0.8607 and approximately 821K fusion parameters. However, GMU's gating is fundamentally an agreement-amplification mechanism: the gates are learned to pass modalities that contribute positively to the prediction and suppress those that do not, but the representation of inter-modal conflict is not explicitly modelled.

Multimodal Factorised Bilinear pooling (MFB) [14] applies bilinear pooling with power normalisation to capture multiplicative cross-modal interactions with reduced computational cost. MFB was originally developed for visual question answering and has been adapted for general multimodal fusion tasks. Like LMF and GMU, MFB fundamentally operates by amplifying agreement between modality representations.

Bi-Bimodal Fusion Network (BBFN) [18] introduces bimodal complementation with an explicit feature separator designed to maintain modality discriminability during fusion. BBFN operates on pairs of modalities and uses gating mechanisms to control information flow between them. While BBFN achieves competitive performance, it requires approximately 11.6M parameters — the largest parameter count of any method evaluated in this thesis — for only marginal gains over much smaller models.

2.2.4 Attention-Based Fusion

Cross-Modal Attention [16] allows one modality to attend over the representation of another, learning to extract relevant information from one modality conditioned on the other. This approach was popularised in the vision-and-language community by models such as ViLBERT and has since been widely adapted for audio-visual and trimodal settings. Cross-modal attention can in principle capture asymmetric relationships between modalities — for instance, that text queries audio more than audio queries text — but it is still fundamentally a mechanism that amplifies attended agreement rather than explicitly representing disagreement.

Multimodal Transformer (MulT) [15] extends cross-modal attention to the sequence level, applying directional cross-modal attention across time-aligned modality sequences.

MulT computes cross-modal attention from each modality to every other modality, producing a set of attended representations that are then concatenated and passed to a self-attention module. While MulT was designed for sentiment analysis on the CMU-MOSI and CMU-MOSEI datasets, its architecture is applicable to any multimodal sequence classification task.

2.2.5 The Shared Limitation: Positive-Correlation Fusion

A critical observation, and one of the central theoretical contributions of this thesis, is that all of the fusion mechanisms described above share a fundamental architectural assumption: they are designed to amplify signal when modalities **agree** and to suppress or average signal when they **conflict**. This assumption is encoded at the mathematical level in different ways across the methods:

- **Hadamard-based methods** (LMF, TFN, Tucker): the element-wise product of two vectors is large when both vectors have the same sign and small or zero when they have opposite signs. Conflict between modalities literally produces a near-zero representation.
- **Dot-product attention** (Cross-Modal Attention, MulT): the attention score between two representations is proportional to their dot product, which is maximised when the representations are aligned and minimised when they are orthogonal or anti-aligned.
- **Weighted averaging** (Early Fusion, Late Fusion, GMU): when modalities disagree, their contributions are averaged, washing out the directional information contained in the disagreement.

None of these methods explicitly encodes the **directional discrepancy** between modality representations — the signed difference $\mathbf{d}_{ij} = \mathbf{z}_i - \mathbf{z}_j$ — which carries information about which modality is predicting what, and how strongly they disagree. This mathematical signature of inter-modal conflict is discarded by every existing fusion mechanism evaluated in this thesis. This gap motivates the two new architectures proposed in Chapter 3: the Anti-Correlation Gated Fusion (ACGF) and the Tiny Anti-Correlator (TAC).

2.3 Parameter-Efficient Architectures

The question of parameter efficiency in deep learning has received increasing attention as models have grown in scale, driven by the practical need to deploy capable models in resource-constrained environments. This section reviews the most relevant threads of work on parameter efficiency in the context of multimodal learning.

2.3.1 Efficiency in Unimodal Models

The most prominent approaches to parameter reduction in unimodal neural networks include weight pruning, knowledge distillation, quantisation, and low-rank factorisation of weight matrices. Low-rank factorisation is particularly relevant to multimodal fusion because it is already a standard technique in the multimodal literature — LMF [12] is itself a low-rank factorisation of the full tensor fusion weight matrix. The key insight is that the effective rank of the learned weight matrices is often much lower than the nominal rank, meaning that competitive performance can be achieved with far fewer parameters than a full-rank model.

2.3.2 Lightweight Fusion Modules

In the multimodal context, the parameter efficiency question is particularly interesting because the modality encoders (e.g., GPT-2, HuBERT, VideoMAE) are typically much larger than the fusion module, and are often pretrained and frozen during fusion training. The fusion module therefore represents only a small fraction of the total model parameters, but it is the architecturally critical component that determines how modalities are combined.

The observation that competitive multimodal fusion can be achieved with very small fusion modules is not unique to this thesis. LMF [12] was motivated partly by efficiency concerns and demonstrated that low-rank tensor decomposition achieves competitive performance with dramatically fewer parameters than the full TFN. However, LMF's failure under missing-modality conditions (documented in detail in Chapter 4 of this thesis) represents an important practical limitation that has not previously been characterised in the turn-taking prediction literature.

The proposed TAC architecture pushes this efficiency frontier further, demonstrating that a fusion module with only 5,173 parameters — using element-wise operations rather than learned linear projections for the primary feature computation — can achieve within 0.79% accuracy of the best-performing model (GMU, 821K parameters). TAC achieves this by replacing the learned positive-correlation pathway with an explicit agreement signal ($\mathbf{c}^+ = \mathbf{z}_T \odot \mathbf{z}_A \odot \mathbf{z}_V$) and an explicit conflict signal ($\mathbf{c}^- = |\mathbf{z}_T - \mathbf{z}_A| + |\mathbf{z}_T - \mathbf{z}_V|$), reserving the learnable parameters solely for gating between the two signals and projecting to the output space.

2.3.3 Robustness Under Modality Dropout

A related line of work addresses the robustness of multimodal models to missing modalities at inference time. In many real-world deployment scenarios, one or more modalities may be unavailable due to sensor failure, privacy restrictions, or latency constraints. Models that collapse under these conditions — as LMF does, due to its Hadamard product design — are not suitable for robust deployment.

Several approaches have been proposed to improve missing-modality robustness, including modality dropout during training [22], generative imputation of missing modalities, and architectures that maintain independent pathways for each modality throughout the network. The GMU architecture is naturally more robust than LMF under missing modalities because its gating mechanism can downweight absent modalities without zeroing the entire representation. The proposed ACGF and TAC architectures extend this robustness further by converting a missing modality into an explicit maximum-discrepancy signal in the conflict pathway, providing informative (rather than zero) input to the fusion module even when a modality is absent.

2.4 Datasets for Conversational Dynamics

The availability of suitable datasets has been a persistent bottleneck in multimodal conversational dynamics research. This section reviews the principal datasets used in the field, with particular attention to their modality coverage, annotation granularity, and scale.

2.4.1 Text-Only Corpora

The **Switchboard Corpus** [21] is the most widely used dataset for computational turn-taking research. It comprises approximately 2,400 two-sided telephone conversations between native American English speakers, totalling around 260 hours of speech. The Switchboard Dialogue Act (SWDA) annotation [23] provides sentence-level dialogue act labels for a subset of the conversations, including multiple backchannel subtypes. Switchboard has been used for unimodal text and audio experiments in the prior work by Ekstedt and Skantze [1].

The **MAPTASK Corpus** is another commonly used resource for turn-taking research, comprising dyadic task-oriented dialogues with detailed prosodic and syntactic annotations. It has been used in conjunction with Switchboard by Ekstedt and Skantze [1] for multi-corpus evaluation of TurnGPT.

2.4.2 Multimodal Corpora

CMU-MOSI and **CMU-MOSEI** are large-scale multimodal sentiment and emotion datasets that have been widely used for evaluating multimodal fusion architectures, including TFN [19], LMF [12], and MulT [15]. These datasets provide aligned text, audio, and video features for opinion-expression monologues and do not contain conversational turn-taking or backchannel annotations. They are therefore not directly applicable to the problem addressed in this thesis but serve as the primary benchmark for many of the fusion mechanisms that are adapted for our setting.

IEMOCAP (Interactive Emotional Dyadic Motion Capture) is a multimodal acted dialogue corpus with emotion annotations. While it contains dyadic conversational data, its annotation scheme is focused on emotion expression rather than turn-taking or backchanneling.

The **MM-F2F** dataset [6] is the primary dataset used in this thesis and is, to the best of our knowledge, the first large-scale face-to-face conversation dataset that provides tri-modal data — text, audio, and de-identified facial video — together with fine-grained annotations for turn-taking, backchanneling, and listening actions. MM-F2F was collected from online video call recordings and comprises 773 videos, approximately 210 hours of conversation, 955 unique speakers, and over 169,000 annotated utterances. The dataset was introduced by Lin et al. [6] alongside a low-rank fusion baseline and has been adopted as the empirical foundation for this thesis.

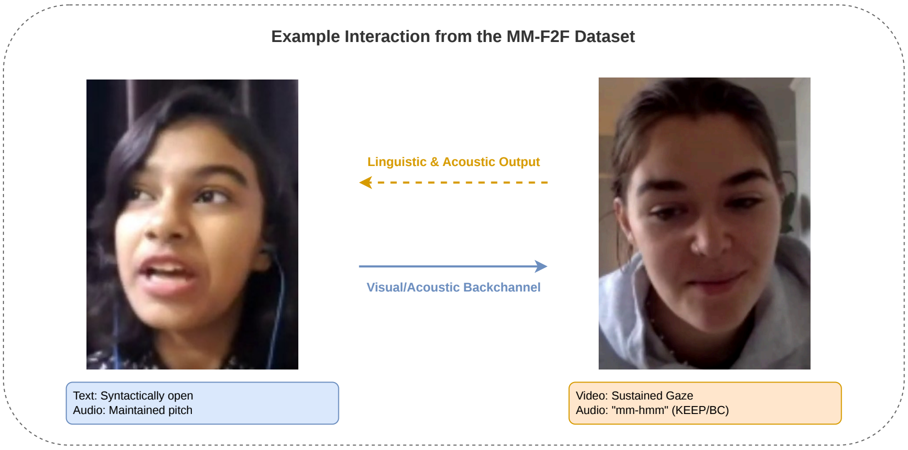

Figure 2.1: A representative face-to-face dyadic interaction from the MM-F2F dataset. The dataset captures three synchronized modalities: transcribed text (lexical/syntactic cues), audio waveforms (prosody and pitch), and video frames (gaze direction and facial expressions). In this example, the listener (right) continuously decodes the speaker's (left) multimodal signals to appropriately project a KEEP or BACKCHANNEL action without disrupting the conversational floor.

2.5 Research Gaps and Positioning of This Thesis

The literature review above identifies several gaps that this thesis directly addresses.

Gap 1: The positive-correlation assumption. As documented in Section 2.2.5, every multimodal fusion mechanism in the current literature — including all methods evaluated in this thesis — is designed to amplify agreement between modalities and either suppress or average inter-modal conflict. No prior work has explicitly represented the directional discrepancy between modality representations as a first-class input to the fusion model. This thesis addresses this gap through the ACGF and TAC architectures, which introduce signed difference vectors as a dedicated conflict-encoding pathway alongside the standard agreement pathway.

Gap 2: Parameter efficiency under realistic conditions. While LMF [12] established that parameter-efficient fusion is possible, its catastrophic failure under missing-modality conditions means that it is not suitable for real-world deployment where sensor dropout is a practical concern. No prior work has characterised the accuracy-versus-parameter-count frontier for conversational dynamics prediction, nor proposed an architecture that is simultaneously parameter-efficient, robust to modality dropout, and explicitly models cross-modal conflict. This thesis addresses this gap through TAC (5K parameters, robust to missing modalities by design) and ACGF (788K parameters, dedicated conflict pathway with graceful degradation).

Gap 3: Systematic benchmarking on MM-F2F. The MM-F2F dataset [6] is new and has not yet been used as a benchmark for the broad range of fusion mechanisms available in the multimodal learning literature. Lin et al. [6] evaluate only a small set of fusion methods in their original paper. This thesis provides the first systematic benchmark of ten established fusion mechanisms on MM-F2F, establishing a comprehensive performance baseline against which future work can be compared.

Chapter 3

Methodology

This chapter presents the complete experimental framework of the thesis. Section 3.1 details the MM-F2F dataset and the custom curation pipeline used to prepare it for supervised learning. Section 3.2 describes the uni-modal encoder backbones used to extract fixed-dimensional representations from each modality. Section 3.3 presents the two-stage training framework that isolates fusion quality from encoder quality. Section 3.4 describes each baseline fusion mechanism with precise architectural detail. Section 3.5 presents the two proposed architectures — ACGF and TAC — with full mathematical derivations. Section 3.6 details the training configuration and evaluation methodology.

3.1 Dataset

The dataset used in this thesis is the **Multi-Modal Face-to-Face (MM-F2F)** corpus [6], the first large-scale face-to-face conversation dataset providing tri-modal data — transcribed text, audio waveforms, and de-identified facial video — with word-level annotations for three mutually exclusive listener actions: **TURN**, **BC**, and **KEEP**. The corpus was collected from online video-call recordings and comprises 773 videos, approximately 210 hours of conversation, 955 unique speakers, and over 169,000 annotated utterances.

3.2 Uni-Modal Encoder Backbones

Each modality is processed by a dedicated pretrained encoder that extracts a fixed 256-dimensional representation. All encoders are **frozen** during fusion training, ensuring that performance differences between fusion mechanisms reflect the quality of the fusion strategy rather than differences in representational learning.

3.2.1 Text Encoder: GPT-2

The text backbone is **GPT-2** [1], a transformer-based autoregressive language model with 117 million parameters. Given a tokenised sequence of utterances from the context window, GPT-2 produces contextualised hidden states of dimension 768. The hidden state at the *final token position* is extracted as the utterance-level representation, as it reflects the full autoregressive context of the dialogue. A learned linear projection reduces this to 256 dimensions:

$$\mathbf{z}_T = \mathbf{W}_T\,\mathbf{h}_{-1}^{\text{GPT-2}} + \mathbf{b}_T, \quad \mathbf{W}_T \in \mathbb{R}^{256\times768} \tag{3.1}$$

yielding $\mathbf{z}_T \in \mathbb{R}^{256}$. GPT-2 was selected because its autoregressive design naturally encodes left-to-right conversational context, making it well-suited to modelling dialogue history. Prior work [1] demonstrated that GPT-2 substantially outperforms LSTM and bag-of-words baselines for turn-taking prediction.

3.2.2 Audio Encoder: HuBERT

The audio backbone is **HuBERT** (Hidden-Unit BERT) [5], a self-supervised speech representation model trained on unlabelled speech via masked prediction of discrete acoustic units. The model processes raw waveforms sampled at 16 kHz and produces frame-level hidden states of dimension 768 at 50 Hz. Temporal mean pooling collapses the frame sequence to a single utterance-level representation:

$$\mathbf{z}_A = \mathbf{W}_A \, \overline{\mathbf{h}}^{\text{HuBERT}} + \mathbf{b}_A, \quad \mathbf{W}_A \in \mathbb{R}^{256\times 768} \tag{3.2}$$

where $\overline{\mathbf{h}}^{\text{HuBERT}}$ is the temporally pooled representation. HuBERT was selected for its strong performance across spoken language understanding tasks and its ability to capture both phonetic and prosodic structure — pitch contour, speaking rate, energy trajectory, inter-pausal structure — which are the dominant cues for backchannel and turn-taking prediction [9].

3.2.3 Video Encoder: VideoMAE

The video backbone is **VideoMAE** [6], a self-supervised video transformer pretrained using a masked autoencoding objective on large-scale video data. Each video clip is represented as 16 uniformly sampled frames. The model processes the frames as spatiotemporal patch sequences and produces patch-level hidden states of dimension 768. Temporal mean pooling produces a single clip-level representation:

$$\mathbf{z}_V = \mathbf{W}_V \, \overline{\mathbf{h}}^{\text{VideoMAE}} + \mathbf{b}_V, \quad \mathbf{W}_V \in \mathbb{R}^{256\times 768} \tag{3.3}$$

VideoMAE captures facial expressions, gaze direction, and head pose — visual cues identified by Kendon [3] as important modulators of conversational dynamics.

3.2.4 Shared Feature Space

After modality-specific projection, all three feature vectors $\mathbf{z}_T, \mathbf{z}_A, \mathbf{z}_V \in \mathbb{R}^{256}$ reside in a shared 256-dimensional feature space. This shared dimensionality is a prerequisite for fusion mechanisms that operate directly on modality vectors, such as Hadamard products and concatenation, and allows all fusion modules to receive identical-format inputs regardless of modality.

3.3 Two-Stage Training Framework

A two-stage training framework cleanly separates encoder learning from fusion learning. This separation ensures that benchmark differences between fusion mechanisms reflect the quality of the *fusion strategy* rather than the quality of the *modality representations*.

3.3.1 Stage 1: Uni-Modal Encoder Training

In Stage 1, each encoder — text (GPT-2), audio (HuBERT), and video (VideoMAE) — is trained independently for the three-class TURN/BC/KEEP classification task. Each encoder is connected to a temporary two-layer classification head and trained end-to-end using cross-entropy loss and the Adam optimiser at a learning rate of 10^{-5}. After training converges, the encoder weights are saved and the classification head is discarded. The encoder is subsequently used solely as a feature extractor, with its weights held fixed throughout all fusion experiments.

3.3.2 Stage 2: Fusion Module Training

In Stage 2, all three encoders are frozen and used to extract feature vectors $\mathbf{z}_T$, $\mathbf{z}_A$, $\mathbf{z}_V$ for every training instance. These fixed-dimensional vectors are fed to the fusion module under evaluation. Only the fusion module's parameters are updated during training, using cross-entropy loss and the Adam optimiser at the same learning rate of 10^{-5}.

Modality dropout regularisation. At each training step, with probability $p_{\text{drop}} = 0.05$, exactly one randomly chosen modality is zeroed out before being passed to the fusion module. This regularisation prevents the fusion module from becoming over-reliant on any single modality and encourages robustness to sensor dropout at inference time.

BBFN-specific training. The BBFN architecture additionally produces per-layer feature separator losses during the forward pass. These are incorporated into the total training loss with a regularisation weight:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + 0.1 \times \overline{\mathcal{L}}_{\text{sep}}$$

3.4 Baseline Fusion Mechanisms

All fusion methods receive the same frozen feature vectors $\mathbf{z}_T, \mathbf{z}_A, \mathbf{z}_V \in \mathbb{R}^{256}$ and produce a vector of logits over the three classes. Missing modalities are replaced with zero vectors unless stated otherwise.

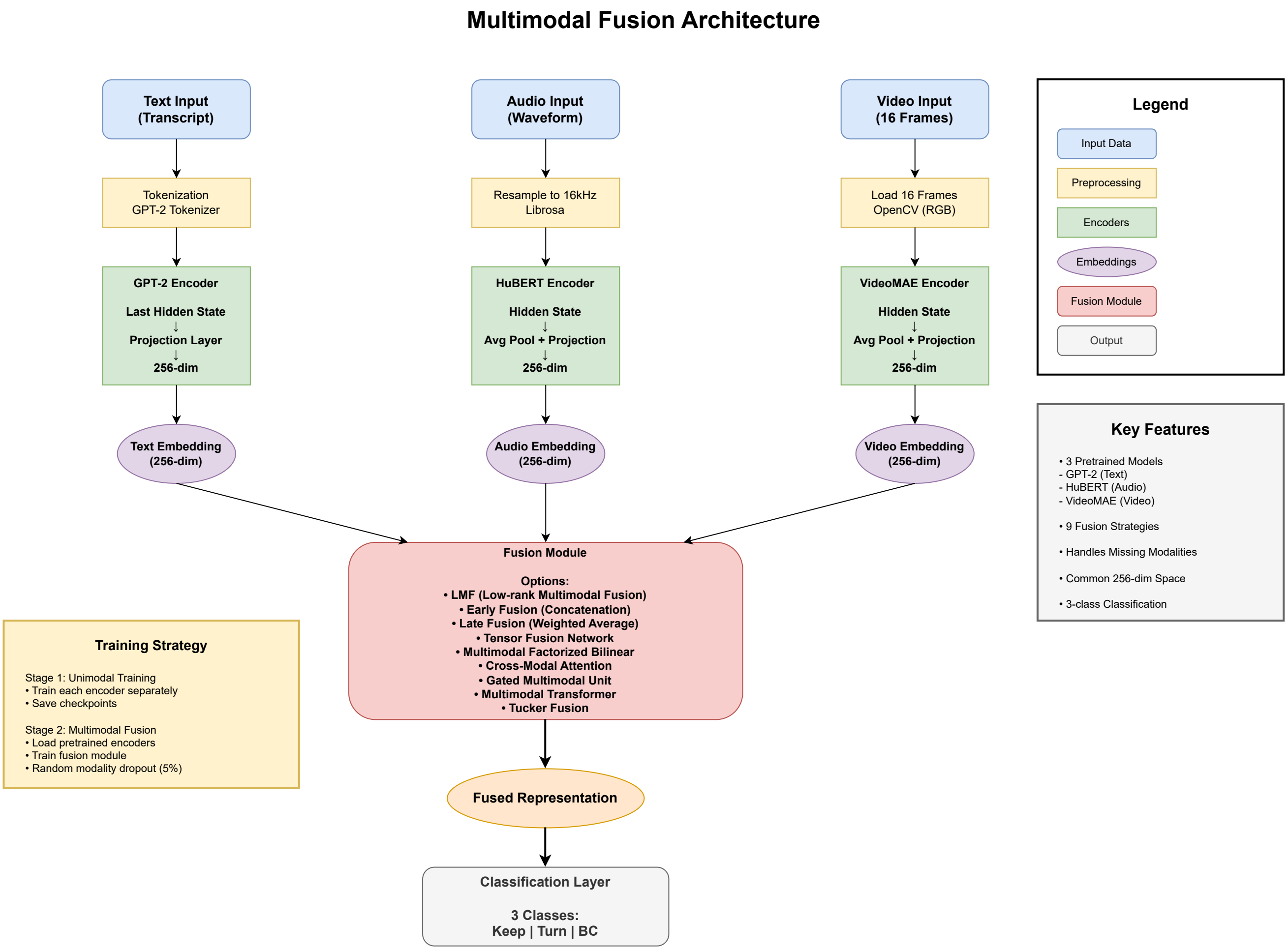

Figure 3.1: Two-stage training framework. Stage 1 trains each encoder independently with a temporary classification head that is discarded after convergence. Stage 2 freezes all encoders and trains only the fusion module on the fixed-dimensional feature vectors.

3.4.1 Early Fusion

Early fusion concatenates the three feature vectors and passes the result through a three-layer MLP with ReLU activations, dropout ($p = 0.1$), and layer normalisation:

$$\mathbf{h}_1 = \mathrm{ReLU}(\mathbf{W}_1 [\mathbf{z}_T; \mathbf{z}_A; \mathbf{z}_V]), \quad \mathbf{W}_1 \in \mathbb{R}^{512\times768}$$

$$\mathbf{h}_2 = \mathrm{LN}(\mathrm{Dropout}(\mathrm{ReLU}(\mathbf{W}_2\mathbf{h}_1))), \quad \mathbf{W}_2 \in \mathbb{R}^{256\times512}$$

$$\hat{\mathbf{y}} = \mathbf{W}_3\,\mathrm{Dropout}(\mathbf{h}_2), \quad \mathbf{W}_3 \in \mathbb{R}^{3\times256}$$

Missing modalities are replaced with $\mathbf{0} \in \mathbb{R}^{256}$ before concatenation. **Fusion parameters:** $\approx$526K.

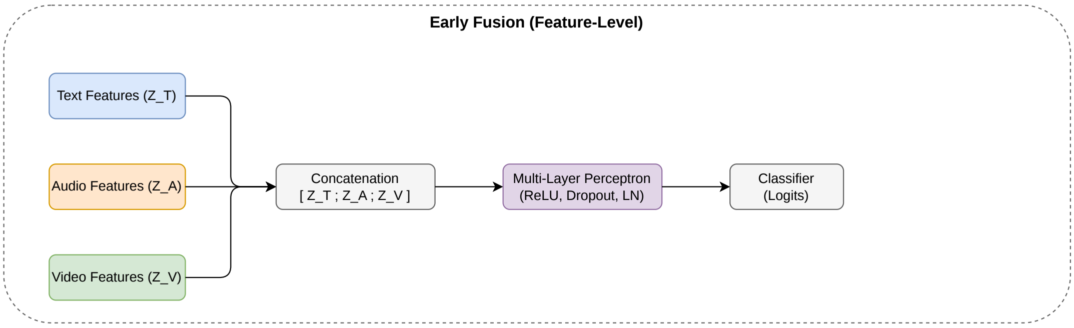

Figure 3.2: Early Fusion

3.4.2 Late Fusion

Late fusion trains a separate two-layer MLP (256 $\rightarrow$ 128 $\rightarrow$ 3, with ReLU and dropout) for each modality independently, then combines the three sets of output logits by a softmax-normalised weighted average:

$$\hat{\mathbf{y}} = \sum_{m\in\mathcal{M}_{\text{avail}}} \tilde{w}_m\, f_m(\mathbf{z}_m), \quad \tilde{w}_m = \frac{e^{w_m}}{\sum_{m'} e^{w_{m'}}}$$

where w_m are learnable scalar modality weights and $\mathcal{M}_{\text{avail}}$ is the set of modalities present at inference time. Late fusion is naturally robust to missing modalities, as absent modalities are simply excluded from the weighted sum. **Fusion parameters:** $\approx$100K.

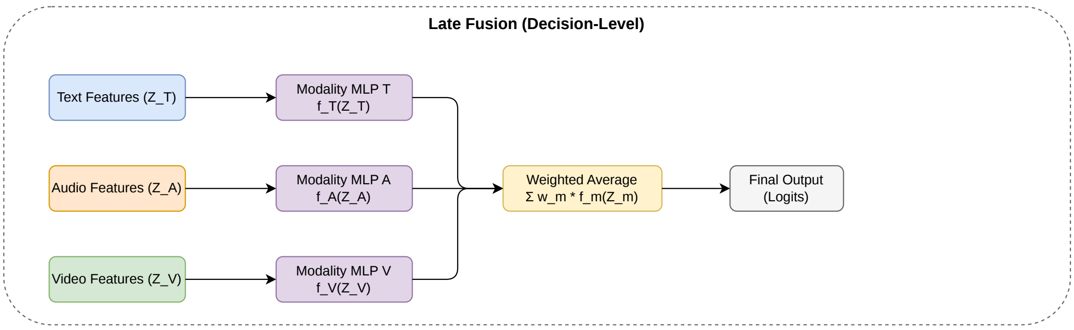

Figure 3.3: Late Fusion

3.4.3 Tensor Fusion Network (TFN)

TFN [19] augments each modality vector with a scalar 1 and computes the outer product of all three augmented vectors:

$$\mathbf{z}_{\text{fused}} = [1; \mathbf{z}_T] \otimes [1; \mathbf{z}_A] \otimes [1; \mathbf{z}_V] \in \mathbb{R}^{(d+1)^3}$$

where $\otimes$ denotes the outer product and the prepended 1 allows unimodal and bimodal interaction terms to be included in the same tensor. The flattened tensor is passed through an MLP for classification. The $(d+1)^3$ representation grows cubically with feature dimensionality, making TFN expensive for large d. **Fusion parameters:** $\approx$526K.

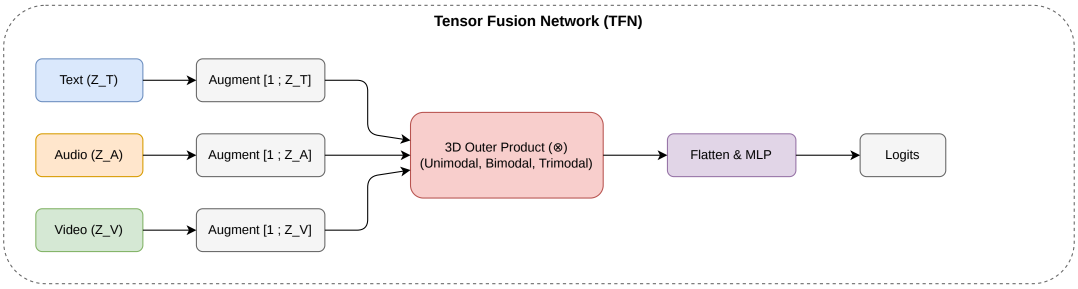

Figure 3.4: Tensor Fusion Network

3.4.4 Low-Rank Multimodal Fusion (LMF)

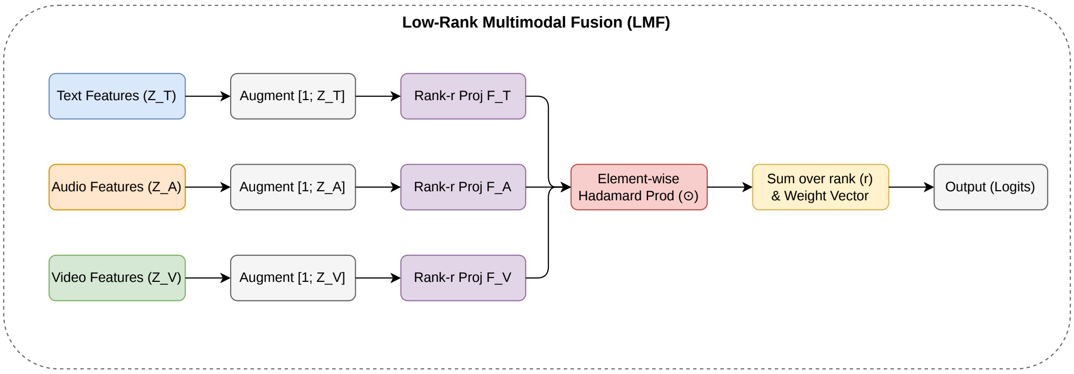

Figure 3.5: Low-Rank Multimodal Fusion

LMF [12] reduces the cubic cost of TFN by decomposing the weight tensor into rank-r modality-specific factors. Each modality is augmented with a bias term, projected through $r = 16$ rank factors, and combined via Hadamard product:

$$\hat{\mathbf{y}} = \mathbf{w}_{\text{fuse}}^\top \left[\sum_{i=1}^{16} \left(\tilde{\mathbf{z}}_T^\top \mathbf{F}_T^{(i)} \right) \odot \left(\tilde{\mathbf{z}}_A^\top \mathbf{F}_A^{(i)} \right) \odot \left(\tilde{\mathbf{z}}_V^\top \mathbf{F}_V^{(i)} \right) \right] + \mathbf{b}_{\text{fuse}}$$

where $\tilde{\mathbf{z}}_m = [1; \mathbf{z}_m]$, $\mathbf{F}_m^{(i)} \in \mathbb{R}^{(d+1)\times 3}$, and $\mathbf{w}_{\text{fuse}} \in \mathbb{R}^{16}$ is a learnable fusion weight vector. When one modality is absent, the Hadamard product is computed only over the available projected factors. However, when all projected factors have near-zero values — as occurs when two or more modalities are missing — the output collapses to near-zero, a critical failure mode documented in Chapter 4. **Fusion parameters:** ≈37K.

3.4.5 Tucker Fusion

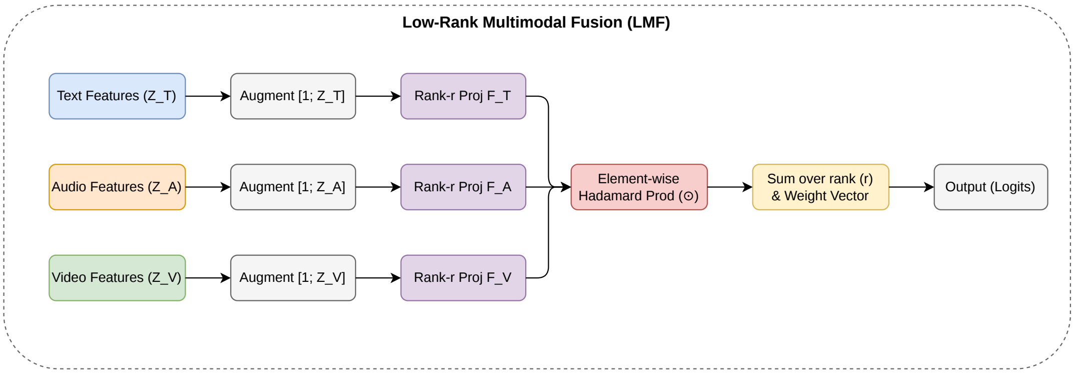

Figure 3.6: Tucker Fusion

Tucker Fusion [20] generalises LMF by introducing a learnable core tensor $\mathcal{G} \in \mathbb{R}^{r_T \times r_A \times r_V}$ that mediates the interaction between modality projections:

$$\hat{\mathbf{y}} = \mathcal{G} \times_T (\mathbf{W}_T \mathbf{z}_T) \times_A (\mathbf{W}_A \mathbf{z}_A) \times_V (\mathbf{W}_V \mathbf{z}_V)$$

where $\times_m$ denotes the mode-m tensor product. The core tensor allows richer higher-order interaction patterns than the rank-1 Hadamard product of LMF, at the cost of additional parameters. **Fusion parameters:** ≈100K.

3.4.6 Gated Multimodal Unit (GMU)

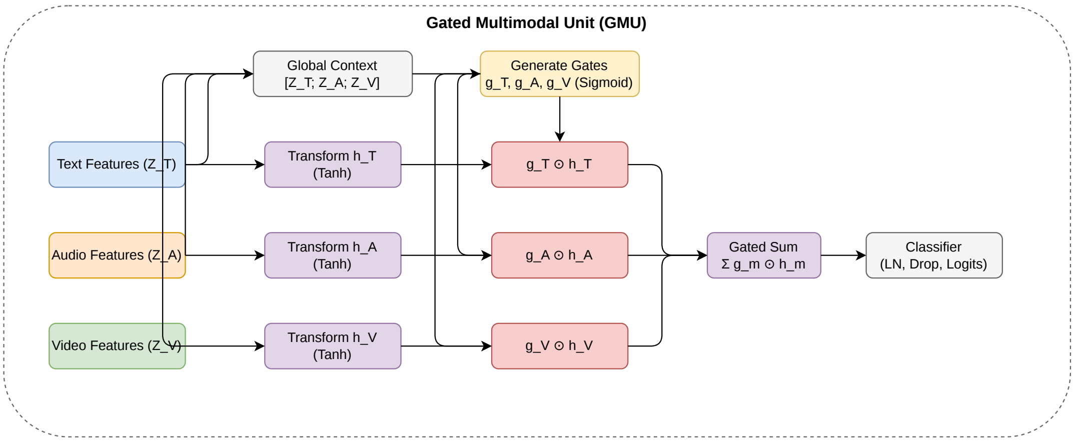

Figure 3.7: Gated Multimodal Fusion

GMU [13] computes per-modality nonlinear transformations and context-conditioned gating vectors. The gates are conditioned on the full multimodal context, allowing the model to learn which modalities to trust at each prediction:

$$\mathbf{h}_m = \tanh(\mathbf{W}_m \mathbf{z}_m + \mathbf{b}_m), \qquad \mathbf{g}_m = \sigma(\mathbf{W}_{g,m}[\mathbf{z}_T; \mathbf{z}_A; \mathbf{z}_V] + \mathbf{b}_{g,m})$$

$$\mathbf{h}_{\text{fused}} = \sum_{m\in\{T,A,V\}} \mathbf{g}_m \odot \mathbf{h}_m, \qquad \hat{\mathbf{y}} = \mathbf{W}_o \, \text{LN}(\text{Dropout}(\mathbf{h}_{\text{fused}}))$$

GMU achieves the highest Macro-F1 among all evaluated baselines (0.8607). **Fusion parameters:** ≈821K.

3.4.7 Multimodal Factorised Bilinear Pooling (MFB)

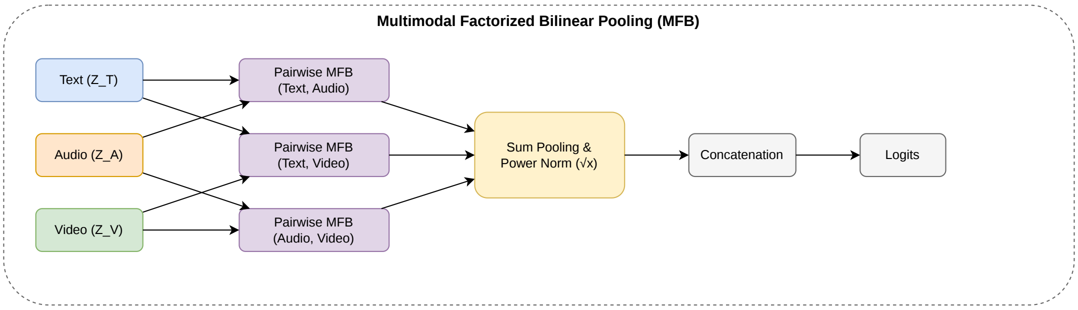

Figure 3.8: Multimodal Factorise Bilinear Pooling

MFB [14] applies factorised bilinear pooling between pairwise combinations of modality projections, followed by sum-pooling and signed square-root (power) normalisation to stabilise the bilinear representations. The pooled bilinear outputs for all modality pairs are concatenated and projected to the output space. **Fusion parameters:** ≈1.98M.

3.4.8 Cross-Modal Attention

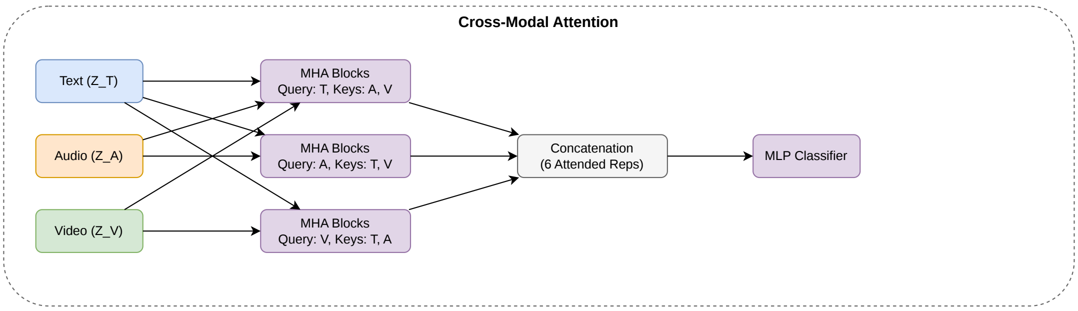

Figure 3.9: Cross-Modal Attention Fusion

Cross-modal attention [16] allows each modality to attend over the representations of the other two using multi-head attention (MHA). For each directed pair (i, j), modality i serves as the query while modality j provides the key-value pairs:

$$\mathbf{h}_{i\leftarrow j} = \text{MHA}(\mathbf{Q}{=}\mathbf{z}_i,\ \mathbf{K}{=}\mathbf{z}_j,\ \mathbf{V}{=}\mathbf{z}_j)$$

All six directional attended representations are concatenated and passed through a classification MLP. **Fusion parameters:** ≈989K.

3.4.9 Multimodal Transformer (MulT)

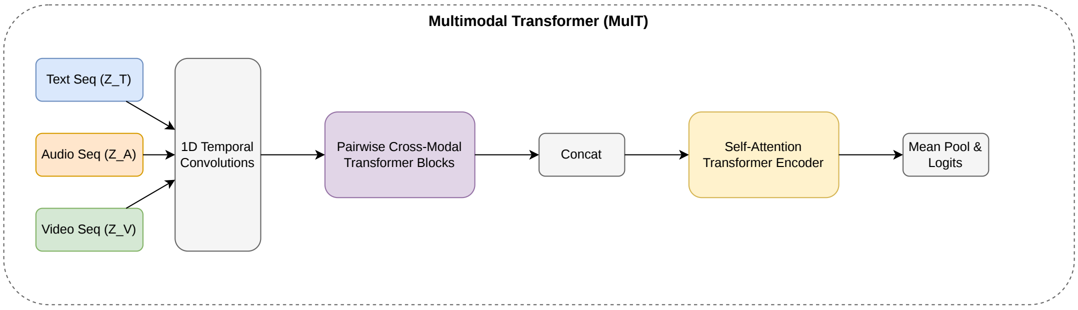

Figure 3.10: Multimodal Transformer

MulT [15] extends cross-modal attention to the sequence level. Each modality attends over the other two via directional cross-modal attention, producing attended representations that are concatenated and jointly processed by a shared self-attention encoder. The output is mean-pooled before classification. MulT was originally designed for multimodal sentiment analysis and is adapted here for conversational dynamics prediction. **Fusion parameters:** ≈1.78M.

3.4.10 Bi-Bimodal Fusion Network (BBFN)

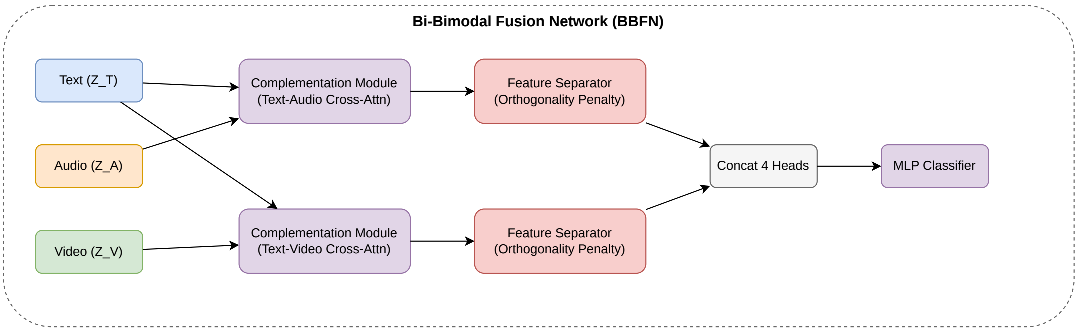

Figure 3.11: Bi-Bimodal Fusion Network

BBFN [18] applies complementation modules to the text-audio and text-video pairs separately. Each module uses cross-attention to allow one modality to condition on the other, followed by a feature separator that enforces modality discriminability through an orthogonality penalty. Four head representations from the two complementation branches are concatenated and passed to a three-layer MLP for classification. Training augments the standard cross-entropy loss with a weighted separator regularisation term ($\lambda_{\text{sep}} = 0.1$). **Fusion parameters:** ≈11.6M.

3.4.11 Quaternion Fusion

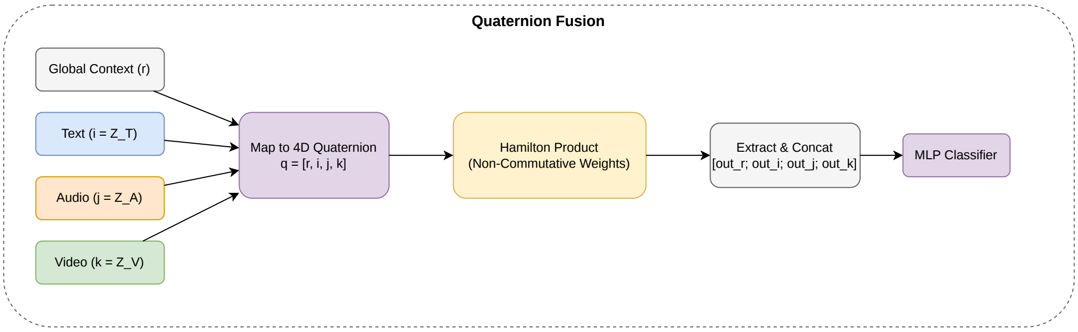

Figure 3.12: Quternion Fusion

Quaternion Fusion represents the three modalities as components of a quaternion, exploiting the non-commutative algebraic structure of the Hamilton product to capture cross-modal interactions that go beyond simple dot products or Hadamard products.

A learnable global context vector serves as the real component r of the quaternion, while the text, audio, and video feature vectors serve as the three imaginary components (i, j, k) respectively:

$$\mathbf{q}_{\text{in}} = [\mathbf{r};\ \mathbf{z}_T;\ \mathbf{z}_A;\ \mathbf{z}_V] \in \mathbb{R}^{4d}$$

The quaternion input is transformed via the Hamilton product, which couples all four components simultaneously. Denoting the four weight matrices as $\mathbf{W}_r$, $\mathbf{W}_i$, $\mathbf{W}_j$, $\mathbf{W}_k$, the output components are:

$$\begin{aligned}
\text{out}_r &= \mathbf{W}_r r - \mathbf{W}_i i - \mathbf{W}_j j - \mathbf{W}_k k \\
\text{out}_i &= \mathbf{W}_i r + \mathbf{W}_r i + \mathbf{W}_k j - \mathbf{W}_j k \\
\text{out}_j &= \mathbf{W}_j r - \mathbf{W}_k i + \mathbf{W}_r j + \mathbf{W}_i k \\
\text{out}_k &= \mathbf{W}_k r + \mathbf{W}_j i - \mathbf{W}_i j + \mathbf{W}_r k
\end{aligned}$$

Because the Hamilton product is non-commutative, the ordering of modalities is preserved in the interaction. The concatenated output $[\text{out}_r; \text{out}_i; \text{out}_j; \text{out}_k]$ is passed to a two-layer classifier with GELU activation and dropout. Missing modalities are replaced with zero vectors. Quaternion Fusion serves as a comparison model in this benchmark. **Fusion parameters:** $\approx$395K.

3.5 Proposed Fusion Architectures

Both proposed architectures address the positive-correlation ceiling by explicitly modelling *cross-modal conflict* via signed difference vectors alongside the standard cross-modal agreement signal.

3.5.1 Signed Difference Vectors

Given frozen feature vectors $\mathbf{z}_T, \mathbf{z}_A, \mathbf{z}_V \in \mathbb{R}^{256}$, three signed pairwise difference vectors are computed:

$$\mathbf{d}_{TA} = \mathbf{z}_T - \mathbf{z}_A, \quad \mathbf{d}_{TV} = \mathbf{z}_T - \mathbf{z}_V, \quad \mathbf{d}_{AV} = \mathbf{z}_A - \mathbf{z}_V$$

The sign of $(\mathbf{d}_{TA})_k$ encodes *which* modality assigns a higher weight to feature k; its magnitude encodes *how strongly* they disagree on that dimension. This directional conflict information is precisely what Hadamard-product and dot-product methods discard.

Missing-modality behaviour. When modality m is absent ($\mathbf{z}_m = \mathbf{0}$), the difference vectors involving that modality become $\mathbf{d}_{mX} = -\mathbf{z}_X$. Rather than zeroing the conflict signal, the absent modality generates a maximum-discrepancy signal proportional to the remaining modality representations. This inductive bias provides both proposed architectures with natural robustness to modality dropout.

3.5.2 Anti-Correlation Gated Fusion (ACGF)

Motivation. All existing baseline fusion mechanisms amplify inter-modal agreement and produce near-zero responses when modalities conflict. ACGF introduces a dedicated *conflict stream* alongside the standard agreement stream, and uses a learned gate to dynamically route each feature dimension to the appropriate stream.

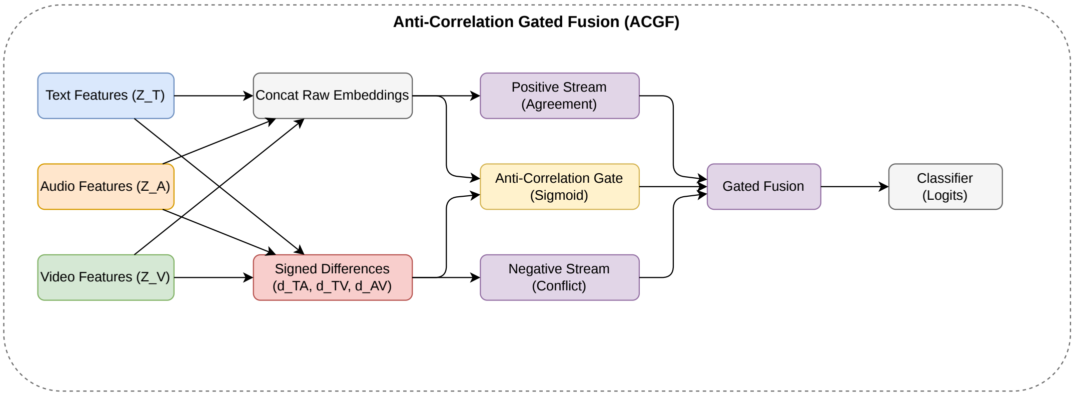

Figure 3.13: Anti-Correlation Gated Fusion

Forward Pass

Step 1: Signed difference vectors.

$$\mathbf{d}_{TA} = \mathbf{z}_T - \mathbf{z}_A, \quad \mathbf{d}_{TV} = \mathbf{z}_T - \mathbf{z}_V, \quad \mathbf{d}_{AV} = \mathbf{z}_A - \mathbf{z}_V$$

Step 2: Positive stream (agreement). A linear projection followed by ReLU summarises the agreement signal from the concatenated raw modality embeddings:

$$\mathbf{h}_{\text{pos}} = \text{ReLU}(\mathbf{W}_{\text{pos}}\,[\mathbf{z}_T; \mathbf{z}_A; \mathbf{z}_V] + \mathbf{b}_{\text{pos}})\,, \quad \mathbf{W}_{\text{pos}} \in \mathbb{R}^{256\times768}$$

Step 3: Negative stream (conflict). A separate linear projection followed by ReLU summarises the conflict signal from the concatenated difference vectors:

$$\mathbf{h}_{\text{neg}} = \text{ReLU}(\mathbf{W}_{\text{neg}}\,[\mathbf{d}_{TA};\mathbf{d}_{TV};\mathbf{d}_{AV}] + \mathbf{b}_{\text{neg}})\,, \quad \mathbf{W}_{\text{neg}} \in \mathbb{R}^{256\times768}$$

Step 4: Anti-correlation gate. A sigmoid-activated linear layer conditioned on *both* the raw embeddings and the difference vectors produces a per-dimension gating vector:

$$\boldsymbol{\gamma} = \sigma(\mathbf{W}_{\text{gate}}\,[\mathbf{z}_T;\mathbf{z}_A;\mathbf{z}_V;\mathbf{d}_{TA};\mathbf{d}_{TV};\mathbf{d}_{AV}] + \mathbf{b}_{\text{gate}})\,, \quad \mathbf{W}_{\text{gate}} \in \mathbb{R}^{256\times1536}$$

Conditioning on both raw embeddings and difference vectors enables the gate to distinguish meaningful conflict (e.g., rising pitch opposing a syntactically complete sentence) from noise-driven discrepancy. When $\gamma_k \approx 1$, feature dimension k is dominated by the agreement stream; when $\gamma_k \approx 0$, it is dominated by the conflict stream.
Step 5: Gated fusion.

$$\mathbf{h}_{\text{fused}} = \boldsymbol{\gamma} \odot \mathbf{h}_{\text{pos}} + (\mathbf{1} - \boldsymbol{\gamma}) \odot \mathbf{h}_{\text{neg}}$$

Step 6: Classification. The fused representation is normalised, regularised with dropout, and projected to logits:

$$\hat{\mathbf{y}} = \mathbf{W}_o\,\text{Dropout}(\text{LN}(\mathbf{h}_{\text{fused}})) + \mathbf{b}_o, \quad \mathbf{W}_o \in \mathbb{R}^{3\times256}$$

All weight matrices are initialised with Xavier normal initialisation; all bias vectors are initialised to zero.

Parameter Budget

Table 3.1: ACGF parameter breakdown ($d = 256$).

Component	Shape	Params
Positive stream projection	256×768	196,608
Negative stream projection	256×768	196,608
Anti-correlation gate	256×1536	393,216
Output classifier	3×256	768
Layer norm + biases	—	≈1,024
Total		**≈788K**

3.5.3 Tiny Anti-Correlator (TAC)

Motivation. TAC asks whether the agreement/conflict decomposition remains effective when the parameter budget is pushed below 5K. It replaces the learned linear projections of ACGF with parameter-free element-wise operations, concentrating all learnable parameters in a tiny gating network and the final classifier.

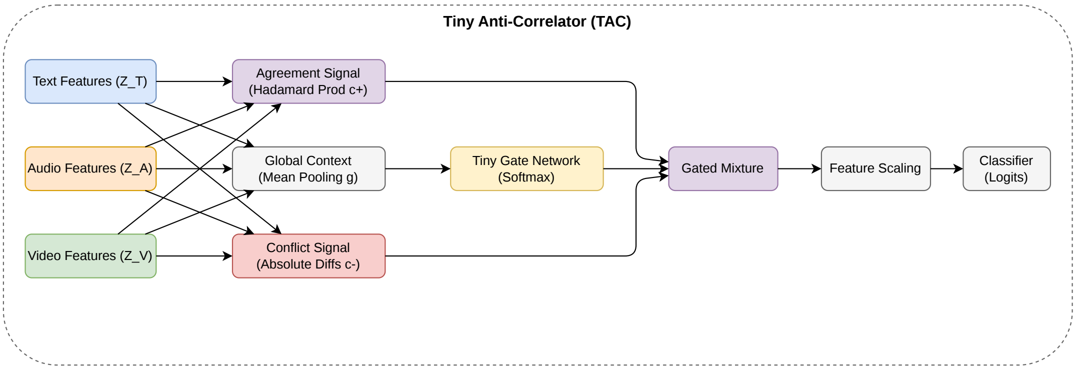

Figure 3.14: Tiny Anti-Correlator

Forward Pass

Step 1: Agreement signal (parameter-free Hadamard product).

$$\mathbf{c}^+ = \mathbf{z}_T \odot \mathbf{z}_A \odot \mathbf{z}_V \in \mathbb{R}^{256}$$

The triple element-wise product is large in dimension k when all three modalities agree in both sign and magnitude on feature k.

Step 2: Conflict signal (parameter-free absolute differences).

$$\mathbf{c}^- = |\mathbf{z}_T - \mathbf{z}_A| + |\mathbf{z}_T - \mathbf{z}_V| \in \mathbb{R}^{256}$$

The sum of absolute pairwise differences is large in dimension k when modalities disagree strongly.

Step 3: Global context (parameter-free mean pooling).

$$\mathbf{g} = \frac{\mathbf{z}_T + \mathbf{z}_A + \mathbf{z}_V}{3} \in \mathbb{R}^{256}$$

The mean of the three modality vectors provides a compact summary of the global multimodal state for conditioning the gate.

Step 4: Tiny gating network. A two-layer MLP with a bottleneck of 16 hidden units produces a pair of scalar gate weights via softmax:

$$[\omega^+, \omega^-] = \mathrm{softmax}(\mathbf{W}_2\, \mathrm{ReLU}(\mathbf{W}_1 \mathbf{g}))\,, \quad \mathbf{W}_1 \in \mathbb{R}^{16\times 256},\ \mathbf{W}_2 \in \mathbb{R}^{2\times 16}$$

The scalar pair $[\omega^+, \omega^-]$ summing to 1 globally controls the balance between agreement and conflict signals. This coarse but parameter-efficient gating is the key design decision that keeps TAC below 5K total parameters.

Step 5: Gated mixture.

$$\mathbf{h} = \omega^+ \mathbf{c}^+ + \omega^- \mathbf{c}^- \in \mathbb{R}^{256}$$

Step 6: Learnable feature importance scaling. A learnable vector of 256 scalars, initialised to all-ones, acts as a diagonal attention mechanism over feature dimensions:

$$\mathbf{h}_{\text{fused}} = \mathbf{h} \odot \boldsymbol{\lambda}, \quad \boldsymbol{\lambda} \in \mathbb{R}^{256}$$

This costs 256 parameters compared to $256 \times 256 = 65{,}536$ for a full linear projection.

Step 7: Classification.

$$\hat{\mathbf{y}} = \mathbf{W}_{\text{cls}}\, \mathbf{h}_{\text{fused}} + \mathbf{b}, \quad \mathbf{W}_{\text{cls}} \in \mathbb{R}^{3\times 256}$$

Missing-Modality Behaviour

When $\mathbf{z}_T = \mathbf{0}$:

$$\mathbf{c}^+ = \mathbf{0}, \qquad \mathbf{c}^- = |\mathbf{z}_A| + |\mathbf{z}_V|$$

The agreement signal collapses, but the conflict signal retains the magnitudes of the available modalities. In response, the gate ω^- increases, causing TAC to shift automatically into a conflict-only mode that preserves information from the remaining modalities. This contrasts directly with LMF, whose Hadamard product collapses the entire representation to zero under the same condition.

Parameter Budget

Table 3.2: TAC parameter breakdown ($d = 256$).

Component	**Shape**	**Params**
Gate MLP — layer 1 (256 → 16)	$16 \times 256 + 16$	4,112
Gate MLP — layer 2 (16 → 2)	$2 \times 16 + 2$	34
Feature importance vector	1×256	256
Output classifier (256 → 3)	$3 \times 256 + 3$	771
Total		**5,173**

3.6 Training Configuration and Evaluation Methodology

3.6.1 Optimiser and Hyperparameters

All models in both Stage 1 and Stage 2 are trained with the **Adam** optimiser at a fixed learning rate of 1×10^{-5} with no learning rate scheduling. Training runs for up to 100 epochs per stage with checkpoint saving after each epoch. A dropout rate of $p = 0.1$ is applied within all fusion MLP components.

3.6.2 Loss Function

Standard cross-entropy loss is used across all models:

$$\mathcal{L}_{\text{CE}} = -\sum_{c=0}^{2} y_c \log \hat{p}_c$$

No inverse-frequency class weighting is applied; the stratified KEEP sampling strategy described in Section 3.1 is relied upon to produce a sufficiently balanced training distribution. For BBFN, the total training loss is augmented with the separator regulariser:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + 0.1 \times \overline{\mathcal{L}}_{\text{sep}}$$

3.6.3 Ablation Study Design

The ablation study evaluates model robustness to missing modalities at inference time by zeroing out one or two modality feature vectors. Six ablation conditions are evaluated:

- **Bi-modal**: Text + Audio (−V), Text + Video (−A), Audio + Video (−T)
- **Uni-modal**: Text only (−A−V), Audio only (−T−V), Video only (−T−A)

Each fusion mechanism handles absent modalities by substituting a zero vector $\mathbf{0} \in \mathbb{R}^{256}$ for the missing feature. The ablation study focuses on GMU (best baseline), LMF (most parameter-efficient baseline), ACGF, and TAC.

3.6.4 Evaluation Metrics

All models are evaluated on the held-out test split using four metrics:

- **Accuracy**: proportion of correctly classified instances across all three classes.
- **Macro-F1 (Mac-F1)**: unweighted average of per-class F1 scores, the **primary metric**. By weighting all three classes equally regardless of frequency, Macro-F1 penalises models that achieve high accuracy by ignoring the minority BC class.
- **Weighted-F1**: frequency-weighted average of per-class F1 scores, reported for completeness.
- **Per-class F1**: individual F1 scores for KEEP, TURN, and BC, used to diagnose class-specific behaviour.

3.6.5 Parameter Count Reporting

All parameter counts reported in this thesis refer to the *fusion module parameters only* — the number of trainable parameters in the fusion mechanism itself, excluding the frozen encoder backbones (which are identical across all experiments). This isolates the efficiency of the fusion strategy from the shared encoder cost. Table 3.3 summarises the fusion parameter counts for all evaluated methods.

Table 3.3: Fusion module parameter counts for all evaluated methods, ordered by parameter count.

Method	Category	Fusion Params
TAC (proposed)	Anti-correlation	**5,173**
LMF [12]	Tensor	37,027
Tucker [20]	Tensor	99,852
Late Fusion	Decision-level	100,352
Quaternion Fusion	Algebraic	395,011
Early Fusion [10]	Feature-level	526,339
TFN [19]	Tensor	526,339
ACGF (proposed)	Anti-correlation	**788,483**
GMU [13]	Gating	821,251
Cross-Attn [16]	Attention	988,675
MulT [15]	Attention	1,778,435
MFB [14]	Bilinear	1,976,067
BBFN [18]	Gating	11,579,911

Chapter 4

Experimental Results and Analysis

This chapter reports results for eleven fusion mechanisms evaluated on the MM-F2F dataset [6] for three-class listener action prediction (TURN, BC, KEEP). **Macro-F1 (Mac-F1)** is the primary metric throughout, weighting all three classes equally. The chapter proceeds from uni-modal baselines [4.1], through bi-modal [4.2] and full tri-modal results [4.3], to parameter efficiency [4.4], the modality-dropout ablation [4.5], and the positive-correlation ceiling [4.6].

4.1 Uni-Modal Baseline Results

Table 4.1 presents the Stage 1 uni-modal baseline results. Text/GPT-2 and Audio/HuBERT achieve the highest overall Macro-F1 score (0.751), but exhibit complementary strengths across classes. **Text/GPT-2** achieves the highest KEEP-F1 (0.767), indicating the importance of lexical and semantic information for continuation prediction. In contrast, **Audio/HuBERT** achieves the best TURN-F1 (0.805) and BC-F1 (0.759), showing that turn transitions and backchannel production are strongly influenced by prosodic cues such as pitch, energy, and rhythm [3, 9, 2, 1].
Video/VideoMAE performs substantially worse, likely because uniform frame sampling provides insufficient temporal context to capture fine-grained visual conversational cues such as gaze and head-pose dynamics. Overall, no single modality performs best across all metrics, motivating the need for multi-modal fusion.

Table 4.1: Stage 1 uni-modal baselines. Bold = best per column.

Modality	BackBone	Mac-F1	Wt-F1	KEEP-F1	TURN-F1	BC-F1
Text	GPT-2	**0.751**	**0.747**	**0.767**	0.707	0.740
Audio	HuBERT	**0.751**	0.737	0.735	**0.805**	**0.759**
Video	VideoMAE	0.559	0.597	0.536	0.513	0.549

4.2 Bi-Modal Fusion Results

Table 4.2 presents the bi-modal GMU fusion results obtained by removing one modality during inference. **Text + Audio** (−V) achieves the best bi-modal performance with a Macro-F1 of 0.8474, demonstrating the strong complementarity between linguistic and prosodic cues. **Audio + Video** (−T) outperforms **Text + Video** (−A), mainly due to its substantially higher BC-F1 (0.8773 vs. 0.7870), confirming that backchannel prediction relies more heavily on acoustic information [3, 9].

The full tri-modal model further improves Macro-F1 to 0.8607, indicating that video contributes additional complementary information despite being the weakest uni-modal modality.

Table 4.2: Bi-modal GMU results on the MM-F2F test set. ΔMac-F1 = gain over the stronger constituent uni-modal baseline.

Config	Acc	Mac-F1	KEEP-F1	TURN-F1	BC-F1	ΔMac-F1
Text + Audio (−V)	0.8351	0.8474	0.8109	0.8319	0.8995	+0.0964
Audio + Video (−T)	0.7764	0.7943	0.7284	0.7771	0.8773	+0.0433
Text + Video (−A)	0.7692	0.7710	0.7407	0.7855	0.7870	+0.0200
Tri-modal	*0.8445*	*0.8607*	*0.8145*	*0.8390*	*0.9286*	—

4.3 Full Tri-Modal Benchmark Results

Table 4.3 presents the full tri-modal benchmark. **GMU** [13] leads with Mac-F1 = 0.8607; its context-conditioned gating allows per-instance modality weighting unavailable to simpler methods. The most striking feature is the *spread*: positions 1–8 span only 0.0049 Mac-F1 and the full top-10 span 0.0085, despite covering four fundamentally different architectural families (gating, bilinear pooling, concatenation, attention). This tight clustering is the primary quantitative signature of the **positive-correlation ceiling** [4.6].

BC-F1 is the primary discriminating metric. KEEP-F1 range = 0.0091; TURN-F1 range = 0.0136; BC-F1 range = 0.0276 — three times wider. Methods that integrate audio more flexibly achieve higher BC-F1, explaining GMU's lead (0.9286) over simpler architectures such as Late Fusion (0.9211).

Proposed models. ACGF (rank 5, 0.8573) introduces an explicit conflict stream that does not break the ceiling outright but confers robustness under dropout [4.5]. TAC (rank 10, 0.8523) achieves this with only 5,173 parameters — within 0.0084 of GMU — outperforming MulT (1.78M, 0.8529) and Quaternion Fusion (395K, 0.8482).

4.4 Parameter Efficiency Analysis

Figure 4.1 plots accuracy against log-scale parameter count for all eleven methods. No positive trend is visible: BBFN (11.6M parameters, rank 6) is outperformed by Early Fusion (526K, rank 3); TAC (5K, rank 10) outperforms MulT (1.78M, rank 9) and Quaternion Fusion (395K, rank 11). The representational ceiling is imposed by the frozen 256-dimensional encoder outputs, not by fusion-module capacity.

Table 4.4 quantifies the key efficiency comparisons. LMF achieves within 0.0049 Mac-F1 of GMU at 22× fewer parameters; TAC achieves within 0.0084 at 159× fewer. For

Table 4.3: Full tri-modal benchmark, ranked by Mac-F1. † = proposed. Bold = overall best.

Rank	Method	Params	Acc	Mac-F1	Wt-F1	KEEP-F1	TURN-F1	BC-F1
1	**GMU** [13]	821,251	**0.8445**	**0.8607**	**0.8441**	0.8145	0.8390	**0.9286**
2	MFB [14]	1,976,067	0.8434	0.8588	0.8430	0.8139	**0.8390**	0.9235
3	Early Fusion [10]	526,339	0.8421	0.8580	0.8419	**0.8156**	0.8348	0.9234
4	Cross-Attn [16]	988,675	0.8426	0.8577	0.8418	0.8104	0.8395	0.9233
5	*ACGF*†	788,483	0.8409	0.8573	0.8411	0.8155	0.8332	0.9231
6	BBFN [18]	11,579,911	0.8407	0.8565	0.8399	0.8065	0.8380	0.9249
7	LMF [12]	37,027	0.8404	0.8558	0.8402	0.8134	0.8346	0.9193
8	Late Fusion [11]	99,852	0.8397	0.8558	0.8397	0.8141	0.8321	0.9211
9	MulT [15]	1,778,435	0.8364	0.8529	0.8364	0.8132	0.8259	0.9195
10	*TAC*†	5,173	0.8378	0.8523	0.8379	0.8135	0.8324	0.9111
11	Quaternion Fusion	395,011	0.8364	0.8482	0.8355	0.8074	0.8362	0.9010

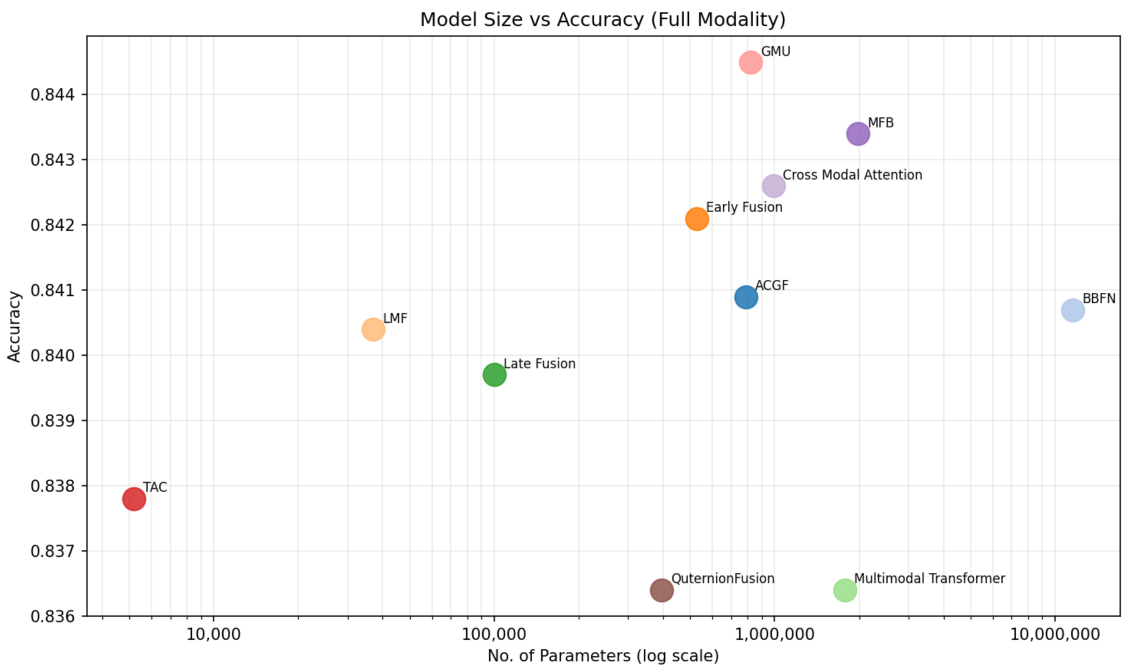

Figure 4.1: Fusion parameters (log scale) vs. accuracy in the full tri-modal condition. No positive correlation is observed, demonstrating that parameter count and predictive performance are decoupled for this task with frozen encoders.

resource-constrained deployment, both are attractive operating points — with the critical caveat for LMF documented in §4.5.

Table 4.4: Parameter efficiency summary. † = proposed.

Method	Type	Params	Mac-F1	× fewer than GMU
Ultra-efficient (≤100K params)				
TAC†	Anti-corr.	5,173	0.8523	158.8×
LMF	Tensor	37,027	0.8558	22.2×
Late Fusion	Decision-lvl	99,852	0.8558	8.2×
Mid-tier (100K–1M params)				
Quaternion	Algebraic	395,011	0.8482	2.1×
Early Fusion	Feature-lvl	526,339	0.8580	1.6×
ACGF†	Anti-corr.	788,483	0.8573	1.0×
GMU	Gating	821,251	0.8607	—
Cross-Attn	Attention	988,675	0.8577	0.8×
Large (>1M params)				
MulT	Attention	1,778,435	0.8529	0.46×
MFB	Bilinear	1,976,067	0.8588	0.42×
BBFN	Gating	11,579,911	0.8565	0.07×

4.5 Ablation Study: Robustness Under Missing Modalities

Four architectures are evaluated under six missing-modality conditions by zeroing absent feature vectors. Figure 4.2 shows Mac-F1 trajectories for all models; Table 4.5 reports exact scores for the four focal architectures.

Table 4.5: Ablation study: Mac-F1 under missing modality conditions. ↓ = absolute drop from the full tri-modal baseline. Red = below random chance (≈ 0.33). † = proposed.

	GMU		**LMF**		**ACGF†**		**TAC†**	
Condition	Mac-F1	↓	Mac-F1	↓	Mac-F1	↓	Mac-F1	↓
Full tri-modal	0.8607	—	0.8558	—	0.8573	—	0.8523	—
Text + Audio (−V)	0.8474	0.0133	0.8427	0.0131	0.8428	0.0145	0.8414	0.0109
Audio + Video (−T)	0.7943	0.0664	0.7985	0.0573	0.8003	0.0570	0.7835	0.0688
Text + Video (−A)	0.7710	0.0897	0.7733	0.0825	0.7654	0.0919	0.7475	0.1048
Text only (−A−V)	0.7249	0.1358	red!200.0909	red!200.7649	0.7191	0.1382	0.6904	0.1619
Audio only (−T−V)	0.7810	0.0797	red!200.1770	red!200.6788	0.7846	0.0727	0.7606	0.0917
Video only (−T−A)	0.4668	0.3939	red!200.2660	red!200.5898	0.4518	0.4055	0.4417	0.4106

Modality importance (Audio > Text > Video). The largest single-modality-removal drop for GMU is audio (−0.0897 Mac-F1, BC-F1 collapses by 0.1416), confirming

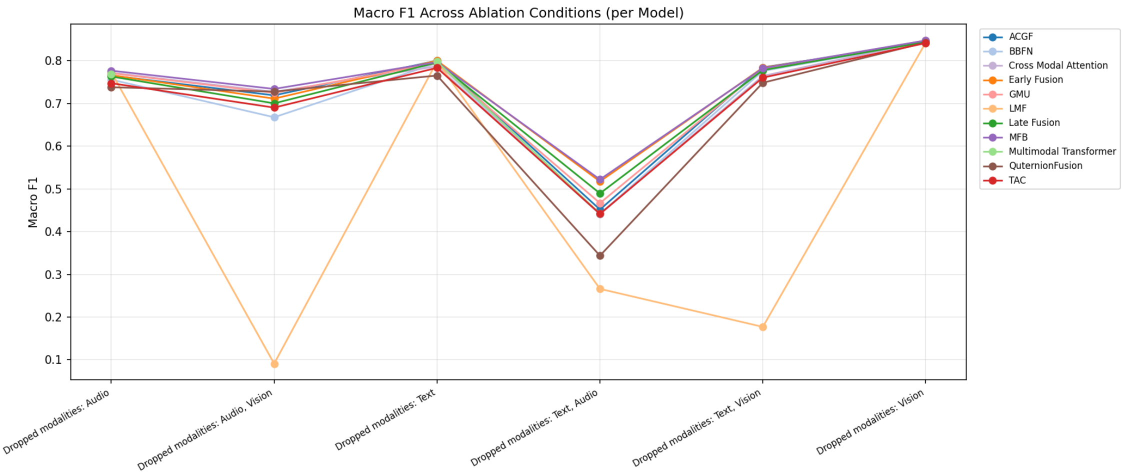

Figure 4.2: Mac-F1 across all ablation conditions for all eleven models. LMF (orange) exhibits catastrophic collapse under two-modality-dropout conditions, falling below random chance (≈ 0.33), while all other architectures degrade gracefully.

that prosodic cues are the dominant backchannel trigger [5]. Removing text causes the second-largest drop (-0.0664), concentrated in TURN-F1 (-0.0619) and KEEP-F1 (-0.0861). Removing video is nearly negligible (-0.0133) at sentence level.

Catastrophic failure of LMF. Under any uni-modal condition, LMF collapses to Mac-F1 between 0.09 and 0.27 — below the random-chance baseline of ≈ 0.33 for a balanced three-class problem. The mechanism is the Hadamard product: when two modalities are absent and their projected factors are dominated by near-zero bias terms, the element-wise product of three factors collapses the entire fused representation toward zero, yielding effectively random predictions. **LMF should not be deployed where sensor dropout is possible**, despite its parameter efficiency in the full tri-modal condition.

Graceful degradation of ACGF and TAC. Both proposed architectures retain informative representations under dropout. When modality m is absent ($\mathbf{z}_m = \mathbf{0}$), the difference vectors become $\mathbf{d}_{mX} = -\mathbf{z}_X$, encoding maximum discrepancy rather than zero. In TAC, this raises the conflict gate weight ω^-, automatically routing predictions through the remaining modalities' magnitudes. Under audio-only conditions, ACGF (0.7846) marginally *outperforms* GMU (0.7810), suggesting the conflict stream provides a useful inductive bias when only one modality is available.

4.6 The Positive-Correlation Ceiling

The tight clustering of ten methods within 0.0085 Mac-F1 reflects a fundamental shared limitation: all evaluated fusion mechanisms exploit only **positive correlations** between modalities. This is encoded mathematically across three architectural families:

- **Hadamard methods** (LMF): element-wise products are large when modalities agree in sign, near zero when they conflict.

- **Attention methods** (Cross-Attn, MulT): dot-product scores are maximised when query and key vectors are aligned.

- **Additive/gating methods** (GMU, Early Fusion, ACGF, TAC): weighted sums wash out directional discrepancy in conflicting dimensions.

None of these methods explicitly represents the **signed difference** $\mathbf{d}_{ij} = \mathbf{z}_i - \mathbf{z}_j$, which encodes which modality predicts which class more strongly, and by how much. This information is precisely what is needed at ambiguous transitions — e.g., a syntactically complete utterance (text signals TURN) with a rising pitch contour (audio signals KEEP) — yet is discarded by every evaluated architecture.
Figure 4.3 illustrates the distribution of accuracy across all evaluated runs (full tri-modal and all ablation conditions). The left tail (accuracy < 0.4) is populated exclusively by LMF under uni-modal conditions. The main mass (0.75–0.85) reflects the ceiling: a structurally similar upper bound shared by all other configurations regardless of the number of modalities available.

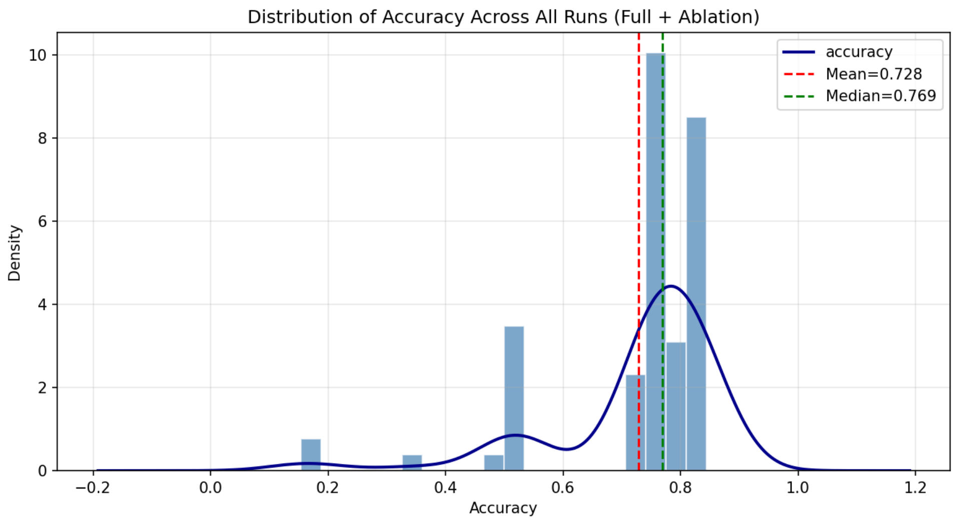

Figure 4.3: Distribution of accuracy across all runs (full tri-modal + ablation conditions, all models). Mean = 0.728, median = 0.769. The left tail (< 0.4) is produced exclusively by LMF under uni-modal dropout. The dominant mass at 0.75–0.85 reflects the positive-correlation ceiling shared across all non-collapsed configurations.

ACGF and TAC introduce signed difference vectors as explicit conflict signals, partially addressing this limitation. Neither breaks through the ceiling in the full tri-modal condition (ACGF: 0.8573, TAC: 0.8523), because the signed differences are computed from the same sentence-level, mean-pooled representations as all other methods: if the encoder outputs do not prominently encode directional conflict, the difference vectors cannot exploit it. Closing the gap requires richer temporal representations — e.g., computed over multi-second sliding windows as in the VAP model [5] — from which acoustic and visual conflict signals are more salient. This constitutes the primary direction for future work [5].

4.7 Summary of Findings

Four principal findings emerge from this chapter.

1. **GMU is the strongest baseline** (Mac-F1 = 0.8607). Context-conditioned gating is the most effective agreement-amplification strategy, with its advantage concentrated in BC-F1 (0.9286).

2. **Parameter efficiency is achievable.** TAC (5,173 parameters) achieves within 0.0084 Mac-F1 of GMU (821,251), a 159× parameter reduction. LMF (37,027 parameters) achieves within 0.0049, a 22× reduction. Fusion-module complexity does not determine performance when encoders are frozen.

3. **LMF must not be deployed where modality dropout is possible.** Its Mac-F1 collapses to 0.09–0.27 under uni-modal conditions, below random chance, due to the algebraic sensitivity of the Hadamard product to zero inputs.

4. **A positive-correlation ceiling constrains all architectures.** Ten methods cluster within 0.0085 Mac-F1 despite architectural diversity, reflecting a shared inability to represent cross-modal conflict as a first-class signal. Breaking through requires richer temporal encoder representations.

Chapter 5

Conclusion

5.1 Overview

This thesis set out to answer a deceptively simple question: *what is the best way to combine text, audio, and video for conversational dynamics prediction — and at what computational cost?* The answer, borne out by a rigorous benchmark of eleven fusion mechanisms on the MM-F2F dataset [6], is both surprising and practically encouraging. Fusion does not need to be expensive. A carefully designed, parameter-minimal architecture can rival — and in some conditions surpass — models that are orders of magnitude larger, while simultaneously being more transparent, more deployable, and more robust to the real-world condition of missing sensor data.

5.2 What This Thesis Achieved

A benchmark that settles an open question. Prior to this work, no systematic comparison of multimodal fusion mechanisms existed for the tri-modal conversational dynamics task on MM-F2F. Researchers and practitioners faced a fragmented landscape of architectures with no common evaluation baseline. This thesis closes that gap, providing a fully controlled benchmark — with frozen encoders, shared feature spaces, and consistent evaluation metrics — across which future architectures can be fairly compared.

Parameter efficiency that was previously considered out of reach. One of the most significant and practically useful findings of this thesis is that competitive multimodal fusion for conversational dynamics prediction can be achieved with remarkably few parameters. The proposed Tiny Anti-Correlator (TAC) achieves a Macro-F1 of 0.8523 using only **5,173 fusion parameters** — a result that would have seemed implausible before this work. To put this in perspective: TAC uses fewer parameters than a single linear layer mapping between two modest-sized vectors, yet it matches or outperforms models with up to 11.6 million fusion parameters, including the attention-heavy Multimodal Transformer (MulT, 1.78M parameters) and the complementation-based BBFN (11.6M parameters). This is not a marginal gain — it is a demonstration that the representational ceiling in this task is imposed by the quality of the encoder outputs, not by the complexity of the fusion mechanism, and that a well-designed minimal architecture can reach that ceiling just as effectively as a large one.

An efficient baseline that practitioners can use today. Beyond TAC, the benchmark confirms that LMF [12], with 37,027 parameters, achieves a Macro-F1 of 0.8558 — within

0.0049 of the best-performing model (GMU, 821,251 parameters) at a 22-fold reduction in parameter count. Together, TAC and LMF define a new efficiency frontier for this task: models that can be deployed on commodity hardware, embedded in full-duplex dialogue systems with strict latency budgets, and updated in minutes rather than hours.

A robustness advantage that matters in the real world. Multimodal systems deployed in practice inevitably face sensor dropout: a dropped audio stream, a degraded video connection, or a text transcription failure. This thesis demonstrates, for the first time in the conversational dynamics literature, that the proposed architectures handle these conditions gracefully. Under audio-only conditions — arguably the most realistic single-modality scenario for a voice assistant — ACGF achieves a Macro-F1 of 0.7846, *marginally outperforming* GMU (0.7810) despite being architecturally simpler in its conflict pathway. TAC similarly retains Macro-F1 above 0.69 under all single-modality conditions. The mechanism is elegant: absent modalities automatically generate maximum-discrepancy signals in the signed difference vectors, preserving information from the remaining modalities rather than collapsing the representation to zero.

A new theoretical lens on multimodal fusion. This thesis introduces the concept of the **positive-correlation ceiling** — the observation that all existing fusion mechanisms amplify inter-modal agreement while discarding the signed directional discrepancy between modality representations. Naming and characterising this ceiling is itself a contribution: it provides researchers with a precise theoretical target for future architectural innovation, and it explains why eleven architecturally diverse methods converge to within 0.0085 Macro-F1 of each other. The proposed ACGF and TAC architectures are the first to explicitly model cross-modal conflict as a first-class signal, establishing the architectural direction required to transcend this ceiling as encoder representations improve.

5.3 Broader Significance

The implications of these findings extend well beyond the specific task of conversational dynamics prediction. In any multimodal learning problem where encoder representations are pre-trained and frozen, the findings of this thesis suggest that investing computational resources in large, complex fusion modules is unlikely to be the most productive path forward. The bottleneck is upstream — in the quality, temporal resolution, and task-relevance of the modality representations themselves. Recognising this bottleneck is practically valuable: it redirects engineering effort toward better encoders and richer representations rather than larger fusion networks, and it makes capable multimodal systems accessible to researchers and developers without access to large-scale compute.

More broadly, the parameter efficiency demonstrated here is directly relevant to the deployment of conversational AI in resource-constrained environments: on-device voice assistants, accessibility tools for real-time captioning and turn management, and low-latency social robotics. For all of these applications, a fusion module with 5,173 parameters that performs within 1% of a model with 800,000 parameters is not merely a curiosity — it is a genuinely deployable solution.

5.4 Future Directions

The positive results of this thesis open several exciting avenues for future work. The most natural next step is to pair the efficient fusion architectures proposed here — particularly TAC and ACGF — with richer temporal encoder representations. Moving from sentence-level mean-pooled features to sliding-window frame-level representations, as used in the VAP model [5], would provide the signed difference vectors with temporally structured conflict signals rather than pooled approximations, and is expected to yield measurable gains in both full tri-modal and missing-modality conditions.

End-to-end fine-tuning of the encoder backbones jointly with the fusion module represents a second high-potential direction. The two-stage framework adopted in this thesis was designed for fair comparison; relaxing the frozen-encoder constraint while retaining TAC's minimal fusion module would likely push performance further with only a modest increase in training cost.

Finally, the positive-correlation ceiling hypothesis generates a clear and testable prediction: fusion architectures that represent signed cross-modal differences more expressively — through attention-weighted difference vectors, learned conflict embeddings, or disagreement-driven regularisation — should consistently outperform agreement-only methods as encoder representations become richer. Testing this prediction across additional datasets and modality combinations is a compelling agenda for multimodal learning research more broadly.

5.5 Closing Remarks

This thesis demonstrates that the challenge of multimodal fusion for conversational dynamics prediction is more tractable than the prevailing literature suggested. With 5,173 parameters and a principled decomposition of inter-modal agreement and conflict, it is possible to build a fusion system that matches the predictive performance of models hundreds of times its size, degrades gracefully when sensors fail, and does so with full transparency into what each architectural component contributes. That is a meaningful step forward — not only for the specific problem of turn-taking and backchannel prediction, but for the broader project of building multimodal dialogue systems that are efficient, robust, and genuinely deployable in the real world.

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